Cognitive Diagrams: Reviewing Categorical Accounts of Linguistic Case

Daniel Ari Friedman
Active Inference Institute
daniel@activeinference.institute
ORCID: 0000-0001-6232-9096
DOI: 10.5281/zenodo.19695259

2026-09-15

Contents

1 Abstract **3**

2 Introduction: Case, Composition, and the Limits of a Shared Diagram **4**
2.1 Contributions and evidence . . . 5

3 Research Questions and Evidence Map **6**

4 Case Systems and Alignment **7**

5 Case-Role Graphs and Categorical Presentations **9**

6 Categorial Grammar and Executable Reductions **12**

7 Case Features and Voice Alternation **14**

8 Compositional Distributional Semantics **17**
8.1 An explicit tensor contraction . . . 17

9 Duality Identities and Diagram Complexity **18**

10 Discourse: Entity Bookkeeping and State-Update Proposals **21**

11 Candidate Similarities and $[0, 1]$-Enrichment **23**

12 Magnitude, Weightings, and Numerical Sensitivity **25**

13 Topos-Theoretic Bridges: Requirements and Present Scope **26**
13.1 What the software represents . . . 26
13.2 What a bridge would require . . . 26

14 Bayesian Case-Role Beliefs and Active-Inference Motivation **28**

15 Cognitive Hypotheses and Testable Comparisons **29**

16 Distributional Utilities and Synthetic Filtering **30**
16.1 Finite score distributions and projection . . . 30
16.2 Pairwise quantile regression and distortion . . . 30
16.3 Filtering, single-factor messages, and factor scores . . . 30
16.4 Caller-defined policy scores . . . 31
16.5 Prediction-error and ERP-inspired proxies . . . 31
16.6 Metrics and reconstruction conventions . . . 33
16.7 Scope and extensions . . . 33
16.8 CEREBRUM as a design vocabulary . . . 33

17 Synthetic Statistical Behavior of the Utilities **34**
17.1 Filtering and transition ablation . . . 34
17.2 Calibration of quantile coverage . . . 34
17.3 Projection and quantile identities . . . 35
17.4 Sensitivity sweeps . . . 35
17.5 Validity controls . . . 35
17.6 Aggregation and localization diagnostics . . . 35
17.7 Interpretation boundaries . . . 36

18 Quantum Diagram Connections and Their Assumptions **37**

19 Case Labels as Measurement Outcomes **38**

20 Proposed Case-Labeled Agent Interfaces **39**

21 Role Policies and the Limits of Type-Based Security **40**
21.1 Requirements for operational enforcement . . . 40
21.2 Limits . . . 41

22 Conclusion **42**
22.1 Next experiments and proofs . . . 42

23 Appendix A: Schematic Construction Panel **43**

24 Appendix B: Notation and Conventions **45**
24.1 Linguistic labels . . . 45
24.2 Algebra and probabilities . . . 45

25 Appendix C: Reproduction and Verification Scope **46**

1 Abstract

Linguistic case offers a setting in which to examine how diagrams connect relational structure, compositional syntax, and uncertainty. This article reviews categorical approaches and supplies an executable collection of deliberately small examples. The implementation includes case-role graphs, pregroup derivations, tensor contractions, synthetic similarity matrices, Bayesian filtering, quantile utilities, and positive-operator-valued measurements. These components share notation, but their mathematical relationships require separate assumptions; a common diagrammatic vocabulary does not establish their equivalence.

The principal contribution is an inspectable comparison of these constructions and their limits. We distinguish generated derivations from illustrative drawings, finite score distributions from Bellman return distributions, presentation statistics from topos invariants, and model-unit prediction errors from physiological measurements. The accompanying suite collects 1909 tests across 94 files; that is a collection count, and a separate source-bound quality receipt records 1909 passing tests with 92.44% line-and-branch coverage (Appendix C). The figure pipeline produces 33 figures from source. All numerical illustrations use synthetic inputs; no corpus study, participant experiment, EEG fit, quantum-hardware run, or deployed-agent security evaluation is reported. Topos-theoretic transfer, cognitive advantages of case diagrams, and case-based authorization remain research proposals with explicit validation requirements. This is working revision 2.7.0; its local validation does not constitute publication or independent mathematical certification.

2 Introduction: Case, Composition, and the Limits of a Shared Diagram

A sentence distinguishes participants and the relations between them. In "Alice chases Bob," changing which participant occupies the subject position changes the interpretation. A representation that stores only an unordered pair of names loses this distinction. Case diagrams make selected relations visible and provide small objects on which to exercise composition, uncertainty, and interpretation.

Three meanings of *case* must remain separate. Morphological case concerns forms and marking; grammatical relations concern positions such as subject and object; semantic roles concern participation in an event. They can correlate without coinciding. NOM is not a universal synonym for agent, and ACC does not always denote a patient. Our examples use role labels as modeling choices, not as a claim that every language has the same inventory.

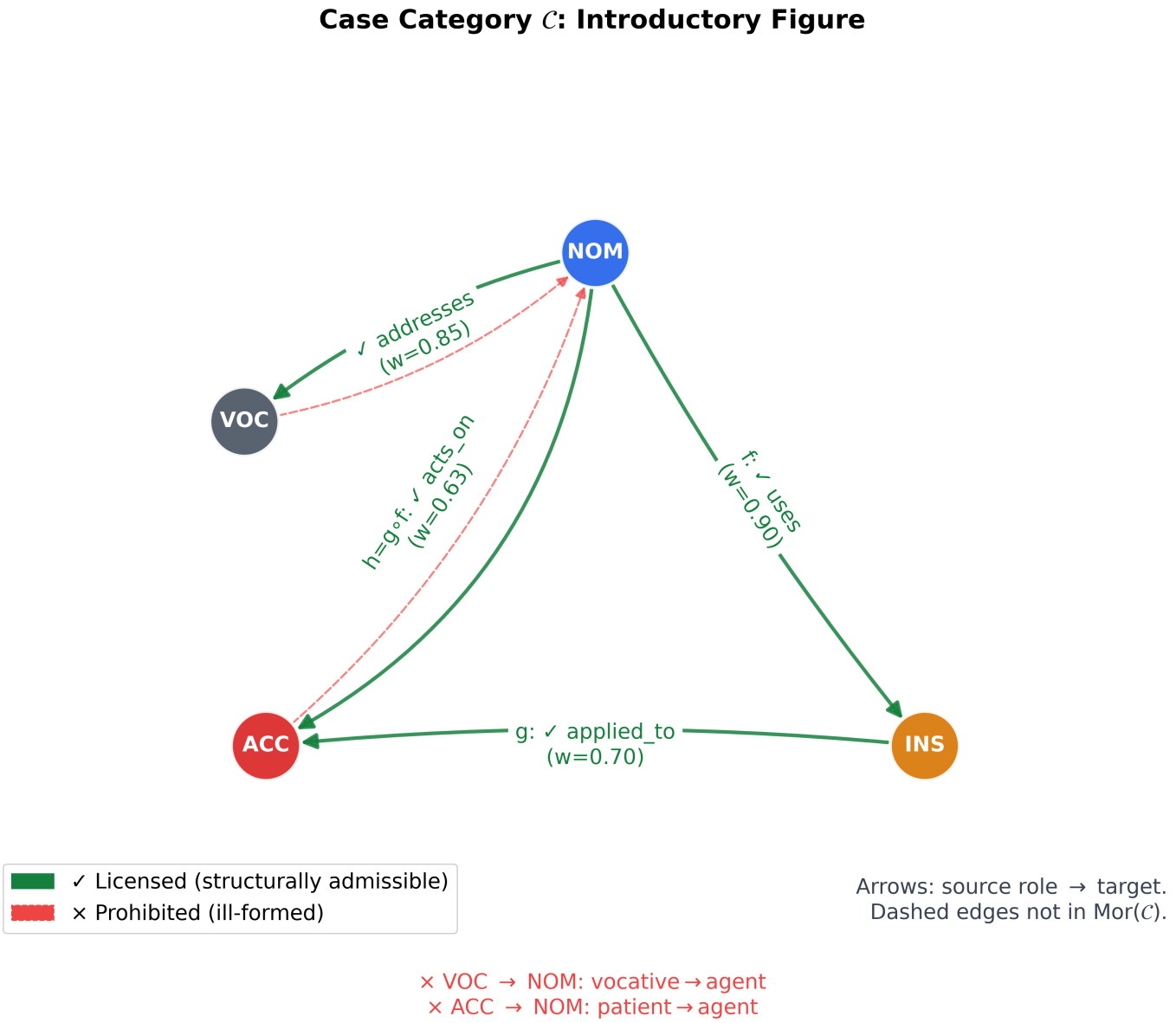


Figure 1: A hand-specified 4-role graph used to introduce the notation. Nodes denote selected role labels and arrows denote named relations. This is a modeling example, not a graph extracted from a corpus.

Categorical compositional distributional semantics provides a mathematical account of how typed grammatical reductions can guide operations on word representations [Coecke et al., 2010]. Enriched-category constructions give a different route from suitable text-extension probabilities to mathematical structure [Bradley et al., 2021]. Neither result implies that an arbitrary role graph, attention matrix, or probability table automatically satisfies the relevant categorical axioms.

Closely related recent work approaches the same post-extraction problem from the opposite direction: how to consolidate unstable extracted claims without overcounting paraphrase or erasing regime and polarity distinctions. Democritus

extracts causal discourse from real text corpora, normalizes it into a categorical presentation, and specifies homotopy-localization over that presentation to consolidate paraphrase families into localized claim classes [Mahadevan, 2026b]. Its quantitative validation concentrates on the upstream extractor, which is benchmarked externally on a frozen causal-extraction split; the downstream localization layer is assessed by artifact audit and internal contrasts, and the paper itself identifies an independent annotation benchmark for its merge, non-merge, and gluing decisions as the most important next step. Our repository supplies the synthetic, claim-ledger-disciplined counterpart to that specification: every normalization analogue, localization analogue, and diagnostic is executable and source-bound, and no numerical example claims corpus evidence. Neither program claims that category theory improves extraction accuracy; both use the categorical language for specification and audit discipline.

2.1 Contributions and evidence

This work contributes a connected exposition and reproducible examples, together with checks that expose where the examples stop. The categorical core distinguishes formal DisCoPy reductions from lightweight role bookkeeping. The numerical core tests normalization, support, composition inequalities, matrix sensitivity, quantile conventions, and measurement-model matrix validity. The manuscript then uses those contracts to delimit proposed cognitive and security interpretations.

The repository contains no trained linguistic model or empirical dataset. Terms such as *prediction*, *precision*, and *return* in compatibility APIs therefore need the operational definitions given below. A synthetic calculation can verify arithmetic under stated assumptions; it cannot establish those assumptions as facts about language or the brain.

The reading order follows a dependency chain: alignment labels; graph and grammar constructions; semantic evaluation; enrichment and magnitude; the distinct requirements of topos theory; filtering and distributional utilities; synthetic statistical behavior of those utilities; measurement analogies; and proposed agent protocols. The research questions in sec. 3 identify which parts are implemented and which require new evidence. The synthetic-statistics section sec. 17 reports seeded computations of the utilities' own behavior and adds no empirical claim.

3 Research Questions and Evidence Map

The project asks what can be represented and checked with small case diagrams. It does not test a universal cognitive theory. The distinction matters because the same drawing can be used as a pedagogical aid, a typed derivation, or a scientific hypothesis, with different standards of evidence.

Table 1: Research questions, evidence, and missing steps.

Question	Present evidence	What remains required
Can selected argument structures be represented compositionally?	Executable pregroup diagrams and tensor contractions	Language-specific grammar coverage and evaluated parsing
Can role similarities form an enriched category?	Explicit axiom checks and max-product closure of a synthetic matrix	Estimated similarities, uncertainty, and justified closure semantics
Can case theories be transferred through a common classifying topos?	Theory-presentation containers only	Geometric theories, sites, equivalence witnesses, invariant transport
Can uncertainty be updated consistently?	Finite Bayesian filtering and distributional utility tests	Learned generative models and held-out prediction
Do diagrams improve human reasoning?	Proposed experimental comparisons	Participants, controlled stimuli, preregistered outcomes
Can case labels help enforce agent authority?	A finite role-policy checker	Authenticated provenance, runtime enforcement, adversarial evaluation
Can post-extraction claim aggregation avoid overcounting paraphrase and erasing regime or polarity distinctions?	Synthetic merge, gluing, and repair diagnostics on constructed claim clusters	Expert-annotated merge, non-merge, and gluing decisions on paraphrastic, partially overlapping, regime-sensitive clusters

The phrase *implemented* below means an operation is available and exercised locally. *Illustrative* denotes a constructed example. *Proposed* denotes an unimplemented or unevaluated connection. Passing software tests does not promote a proposal to a theorem or an empirical finding. Repo-level claim statements in sec. 25 and in the canonical ledger `docs/claim_ledger.md` carry their own status vocabulary: verified, partially_supported, unsupported, and ambiguous / proposed, with the qualified variants verified (synthetic only), verified via receipt, unsupported for raw input, and not_asserted_here used where the cited row warrants them.

The aggregation row follows the benchmark structure that recent causal-discourse extraction work states as its own open step: an independent post-extraction benchmark with expert-annotated merge, non-merge, and gluing decisions [Mahadevan, 2026b]. The synthetic lane rehearses the decision structure; only annotation on real clusters would test it.

4 Case Systems and Alignment

Comparative case analysis separates argument roles from their marking. We use S for the sole argument of a canonical intransitive predicate, A for the more agent-like argument of a canonical transitive predicate, and P for its more patient-like argument. Alignment describes how a specified marking system groups these roles. Case marking of full noun phrases can differ within a language [Comrie, 2013a]; parallel differences hold for pronouns [Comrie, 2013b] and verbal person marking [Siewierska, 2013].

Table 2: Schematic alignment patterns. These are not an exhaustive classification of language-wide behavior.

Pattern	Grouping in the simplified representation
Nominative–accusative	S and A together; P separate
Ergative–absolutive	S and P together; A separate
Tripartite	S, A, and P distinct
Neutral	S, A, and P marked alike
Active–stative / split-S	Subclasses of S pattern differently
Fluid-S	Some S marking varies with construal or context

Historical accounts of semantic participation, deep case, dependency, and proto-roles motivate different choices of analytical unit. They should not be treated as interchangeable theories simply because a diagram can display their labels. In particular, a morphological case system does not determine a unique category of semantic relations.

The software's standard inventory is NOM, ACC, GEN, DAT, INS, LOC, ABL, VOC. ERG, ABS, S, A, and P are additional enum labels used in alignment examples, beyond the 8 standard members. The standard inventory is a convenience, not a universal eight-case ontology. English names in the examples carry assigned grammatical-role metadata; most English full noun phrases do not inflect for nominative versus accusative case.

A context probability in the Fluid-S API specifies a mixture between two case-label assignments. The numerical mixture is well-defined once its endpoints are supplied. Establishing that this mixture describes a language would require documented constructions, annotator agreement, and predictive evaluation. Language names in legacy convenience functions identify the motivating analogy rather than a validated grammar.

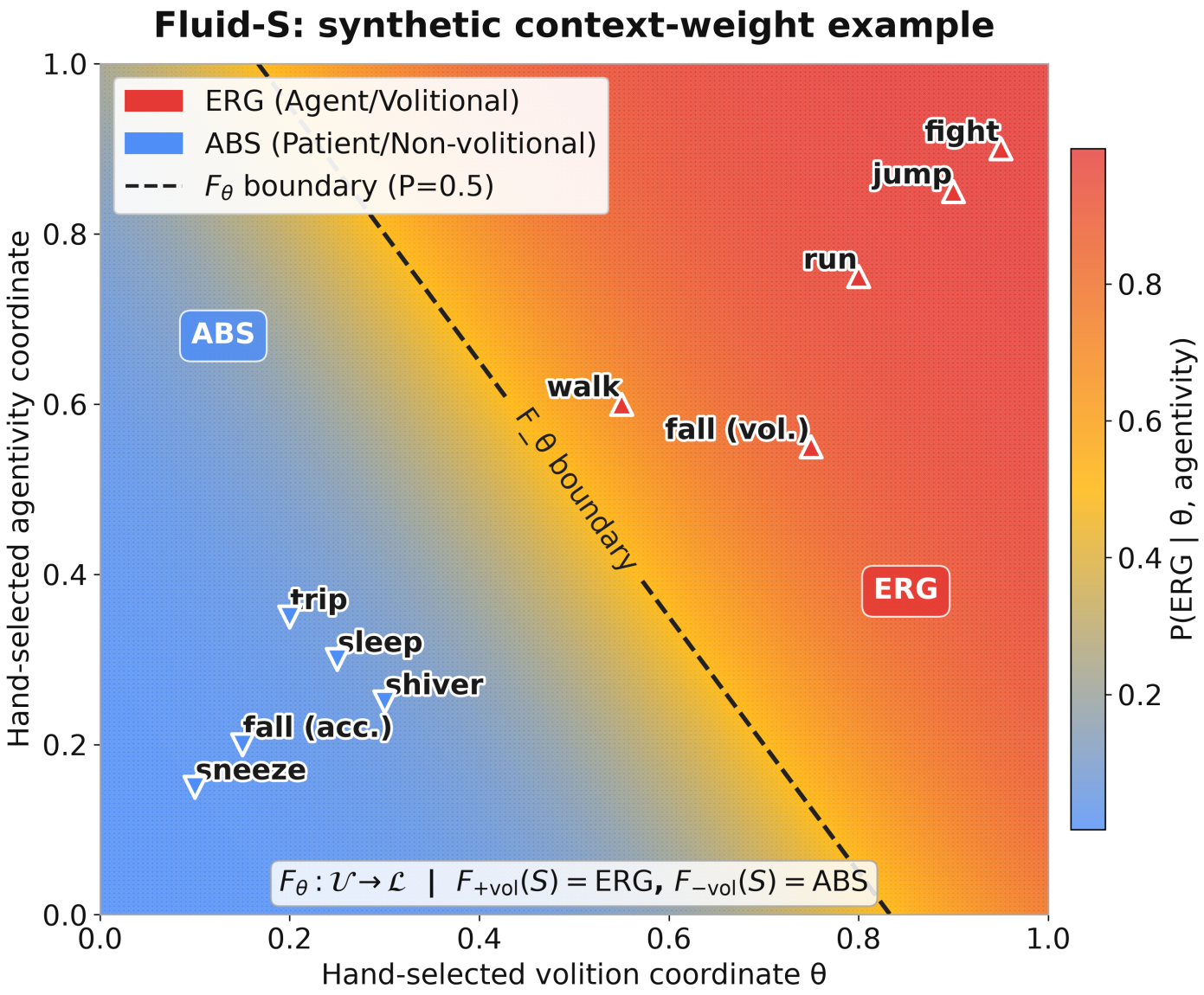


Figure 2: Synthetic Fluid-S weight sweep. The plotted probabilities are supplied by the example generator and illustrate a context-dependent mapping; they are not estimated volitionality judgments or language-specific frequencies.

5 Case-Role Graphs and Categorical Presentations

The Python `CaseCategory` stores a finite role set and a list of labelled arrows. An arrow has source, target, label, and a weight in $[0, 1]$. Identity arrows and composite arrows can be constructed. For composable $f : A \to B$ and $g : B \to C$, the implementation multiplies weights:

$$w(g \circ f) = w(g)w(f). \tag{1}$$

This gives a useful presentation of relational paths. The stored generator list need not contain every composite. Endpoint and weight checks therefore should not be described as a complete implementation of arbitrary hom-sets and their equations. `is_well_formed()` checks the stored representation; it does not prove that a proposed linguistic analysis is adequate.

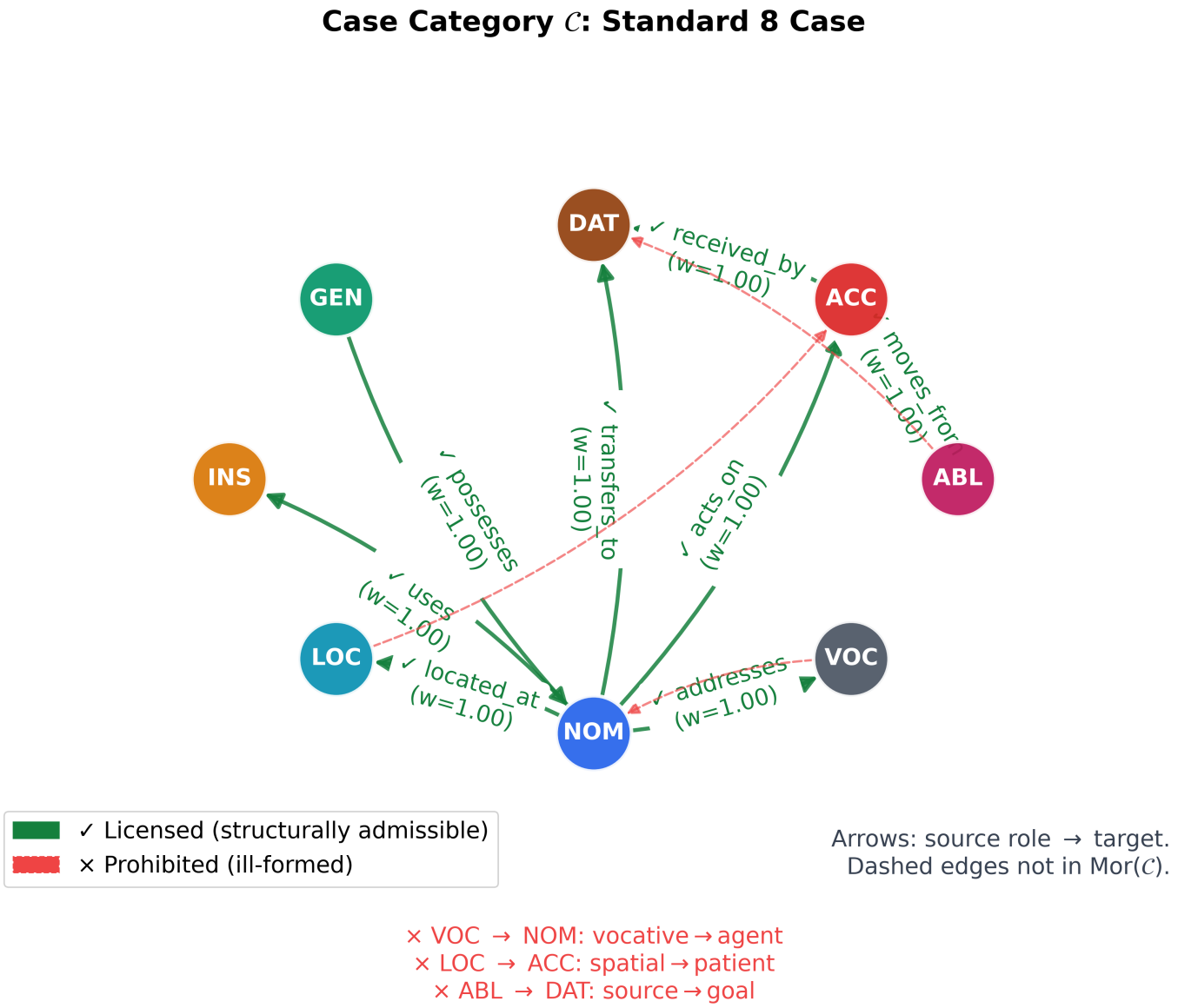


Figure 3: The standard synthetic role graph. Its arrows are explicit modeling choices; their presence is not a measured grammatical universal.

An alignment map assigns source labels to target labels. The two common groupings of S, A, and P differ as maps on named arguments, even when their unlabelled quotient sets have the same cardinality. In particular, NOM alone cannot determine whether a source argument was S or A, so an accusative-to-ergative mapping cannot generally recover the original argument from its surface case.

`AlignmentFunctor` is a lightweight object/arrow mapping helper. Its identity and composition predicates compare endpoints and weights; they do not verify all target-hom membership or labelled-arrow equations. Some alignment factories contain

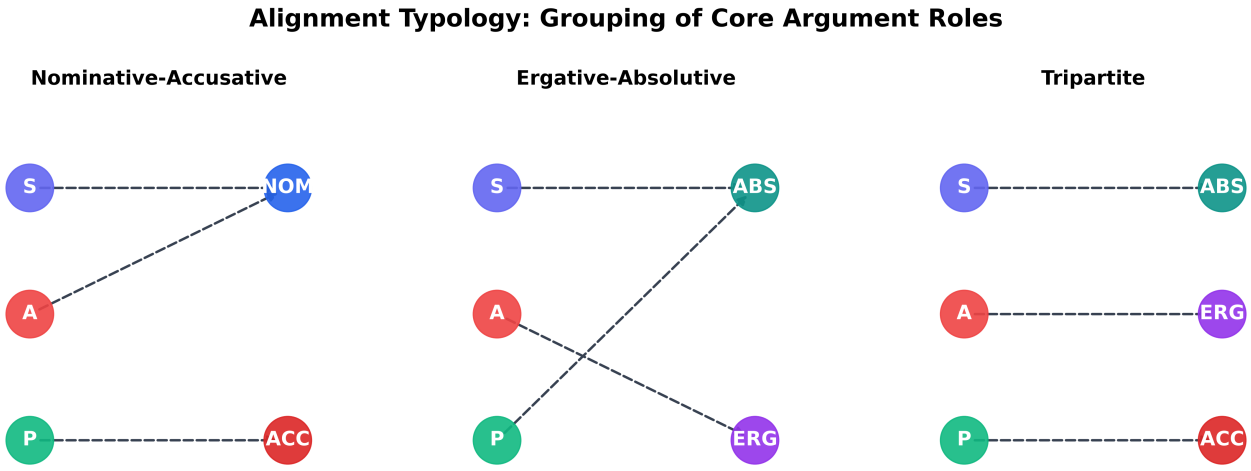


Figure 4: Comparison of schematic alignment maps. The grouped labels illustrate which distinctions are retained or lost; the figure is not an equivalence proof between language grammars.

only objects, so tests over their stored nonidentity arrows can be vacuous. A mathematical functor claim requires an explicit source and target category and preservation of their defining relations.

For actual functors $F, G : C \to D$, a natural transformation requires components $\eta_A : F(A) \to G(A)$ with

$$G(f) \circ \eta_A = \eta_B \circ F(f). \tag{2}$$

The natural-transformation helper checks its supplied components and finite source arrows using the project's representation. This is useful for examples, but neither a dative alternation nor a cross-linguistic correspondence becomes a natural transformation without the required maps and equations.

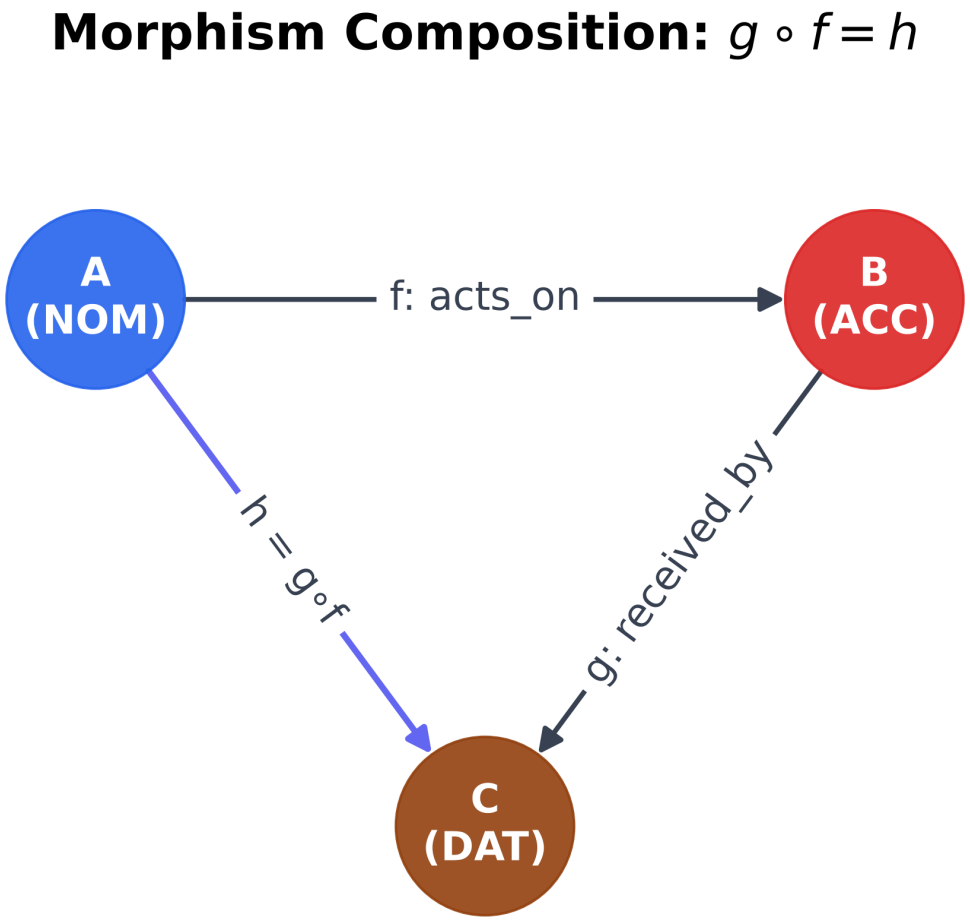


Figure 5: A composition triangle showing the source, intermediate, and target of a selected path. Equality of two paths is an additional relation to establish, not a consequence of drawing a triangle.

6 Categorial Grammar and Executable Reductions

Categorial grammars associate lexical expressions with types that determine how they combine. In a pregroup presentation, a noun type n has left and right adjoints. Ordered reductions use $n^l n \leq 1$ and $nn^r \leq 1$, together with the corresponding expansion inequalities [Coecke et al., 2010]. These are directional operations; arbitrary wire exchange is not an axiom of a nonsymmetric pregroup.

For the hand-assigned types of "Alice chases Bob," the reduction is

$$n\,(n^r s n^l)\,n \longrightarrow s. \tag{3}$$

The implementation builds words and cups with DisCoPy. The assertion below verifies the codomain of this specified derivation. It does not discover lexical types or establish grammaticality independently of them.

```
from discopy.rigid import Ty, Box, Cup, Id
n, s = Ty("n"), Ty("s")
words = (Box("Alice", Ty(), n)
         @ Box("chases", Ty(), n.r @ s @ n.l)
         @ Box("Bob", Ty(), n))
diagram = words >> (Cup(n, n.r) @ Id(s) @ Cup(n.l, n))
assert diagram.cod == s
```

String diagrams represent equations within a specified categorical calculus. A drawing is a proof only when the allowed rewrites and interpretation have been established. Visual similarity alone does not prove a linguistic universal, semantic equivalence, or independence from word order.

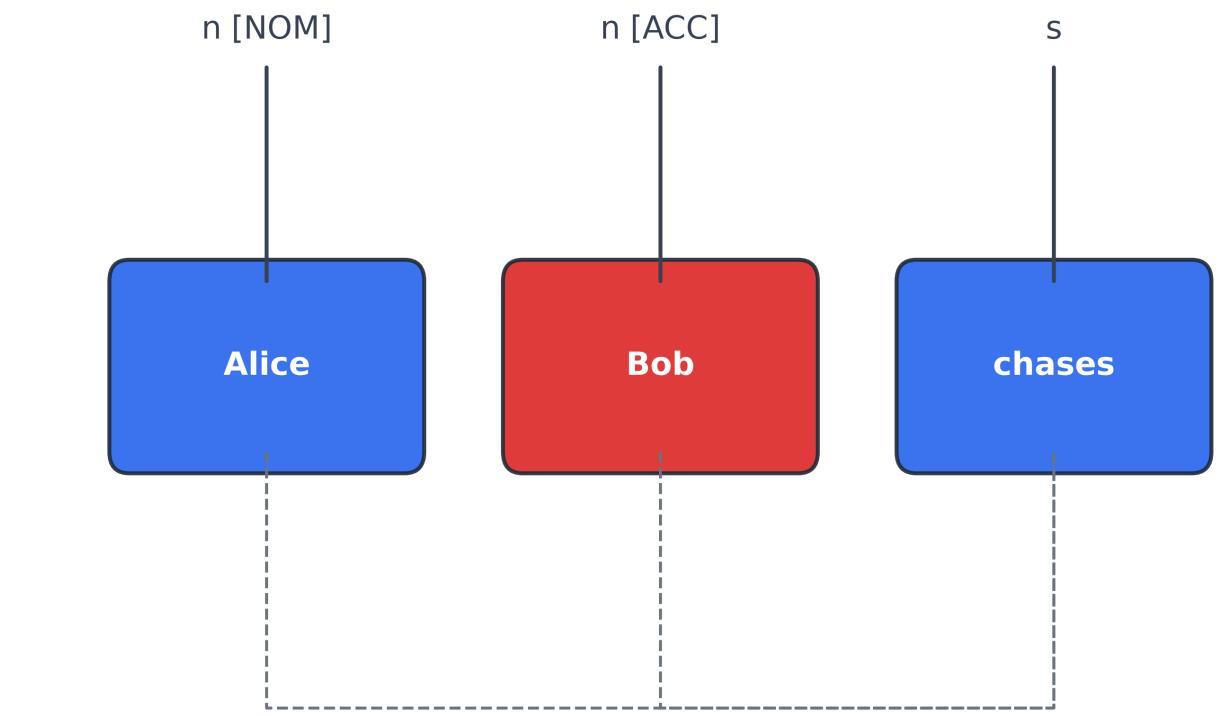


Figure 6: Native rendering of the transitive example. Case colors are supplied metadata; the drawing is a schematic companion to the executable DisCoPy construction.

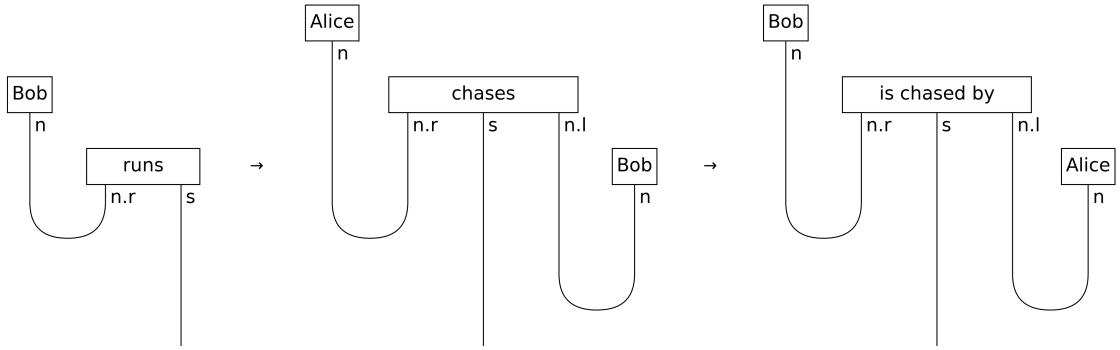


Figure 7: Intransitive, transitive, and passive examples with explicitly assigned lexical types. Cup counts describe these constructions rather than a universal complexity scale for voice.

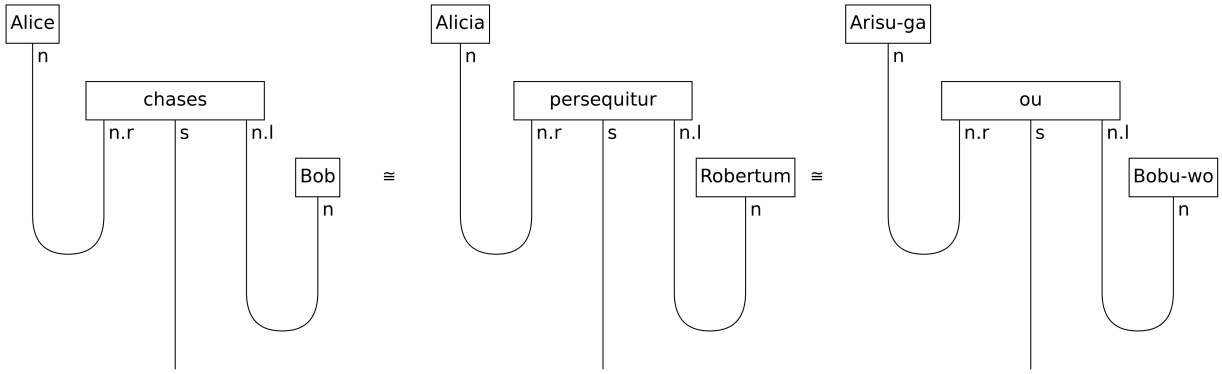


Figure 8: Lexical substitutions in a shared normalized diagram template. Language labels do not imply that the displayed order is each language's surface order or that the code implements those languages' grammars.

7 Case Features and Voice Alternation

A case label attached to a drawing is different from a case-refined type enforced by a parser. Most constructors in this project use the same noun object `Ty("n")` and store NOM/ACC distinctions as metadata. They therefore do not reject every mismatched case feature.

A proposed refinement can introduce distinct objects n_{NOM} and n_{ACC} and type a transitive verb as $n^r_{\mathrm{NOM}} s n^l_{\mathrm{ACC}}$. Then the intended reductions involve those exact objects. This requires corresponding lexical assignments and explicit coercions where a grammar licenses them; it is not achieved by decorating an unrefined diagram after construction.

Passivization changes grammatical realization while preserving selected semantic participation. The project offers two schematic passive templates: a compact-closure construction and a Word-based construction. Neither performs a type permutation; a Swap is available only in an appropriate symmetric extension and is not itself a general linguistic derivation of passive voice. Nor does changing a participant's grammatical role necessarily change that participant's identity.

Proof–type correspondences can clarify what a grammar derivation establishes. They do not make arbitrary case assignment computationally identical to Hindley–Milner inference, nor do they guarantee native-speaker judgments. This implementation performs selected typed compositions and role bookkeeping; it does not implement general case-feature inference, a full Lambek prover, or monadic root syntax.

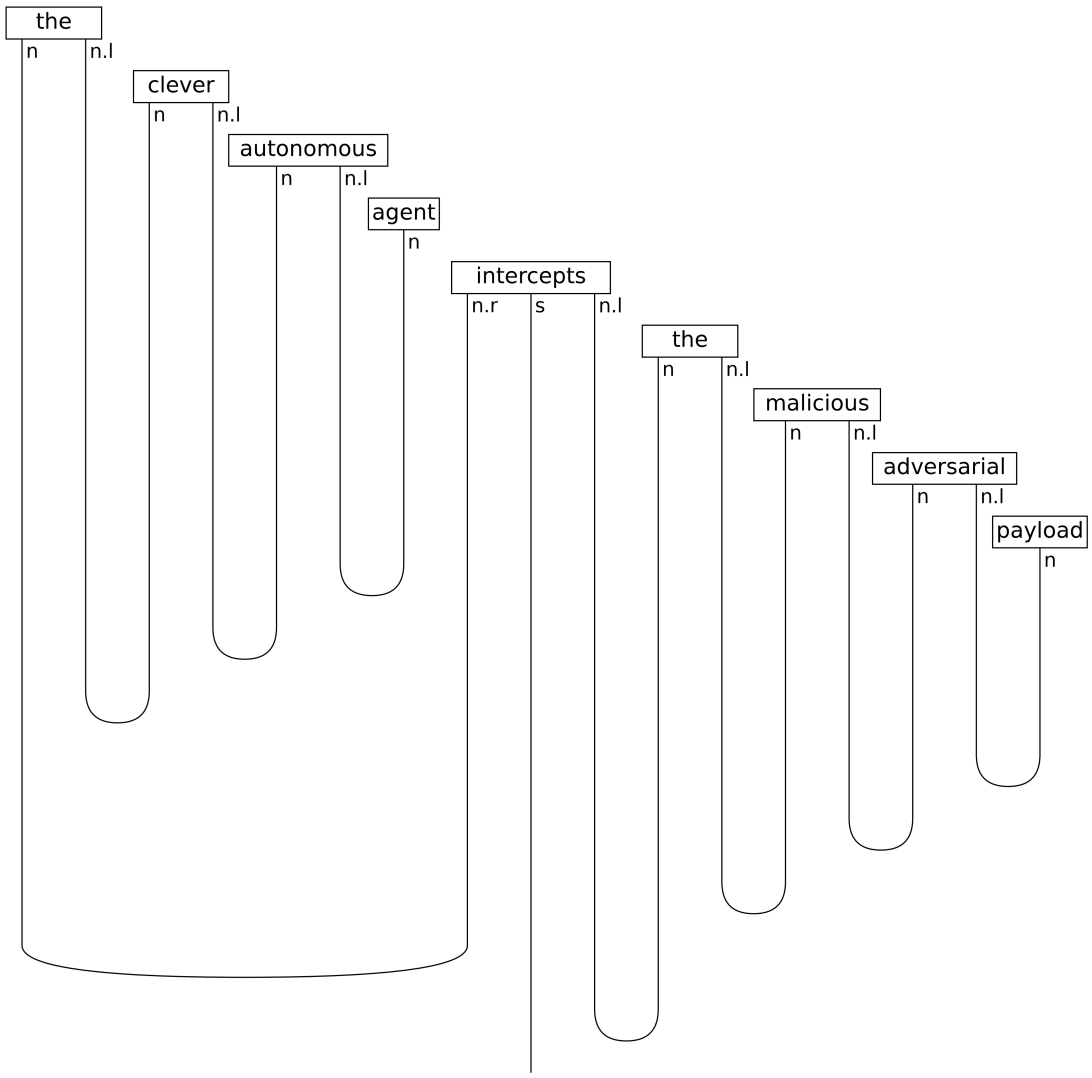


Figure 9: A larger executable DisCoPy example using assigned modifier and argument types. Successful contraction checks the supplied type sequence.

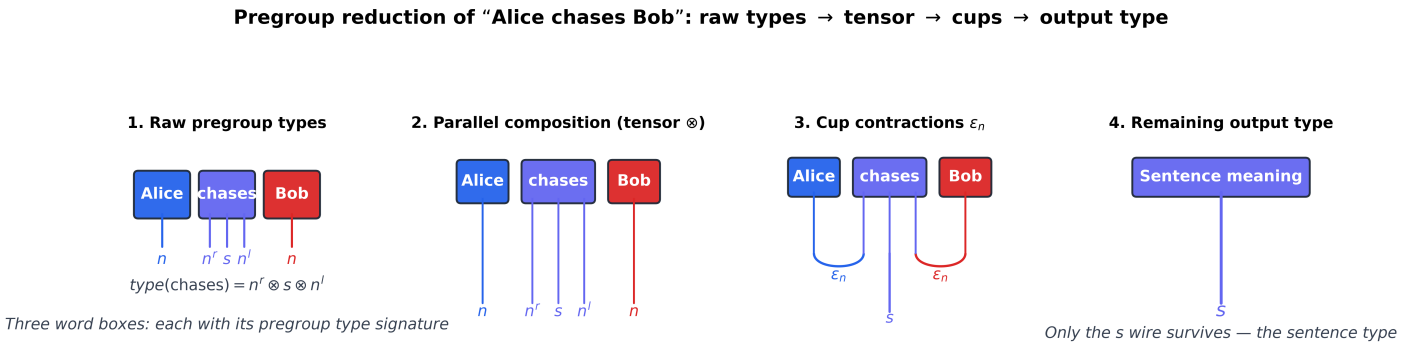


Figure 10: Pedagogical unpacking of the transitive reduction: lexical types, juxtaposition, cup contraction, and the remaining sentence output. The reduced output type is s; the semantic sentence morphism is not thereby erased.

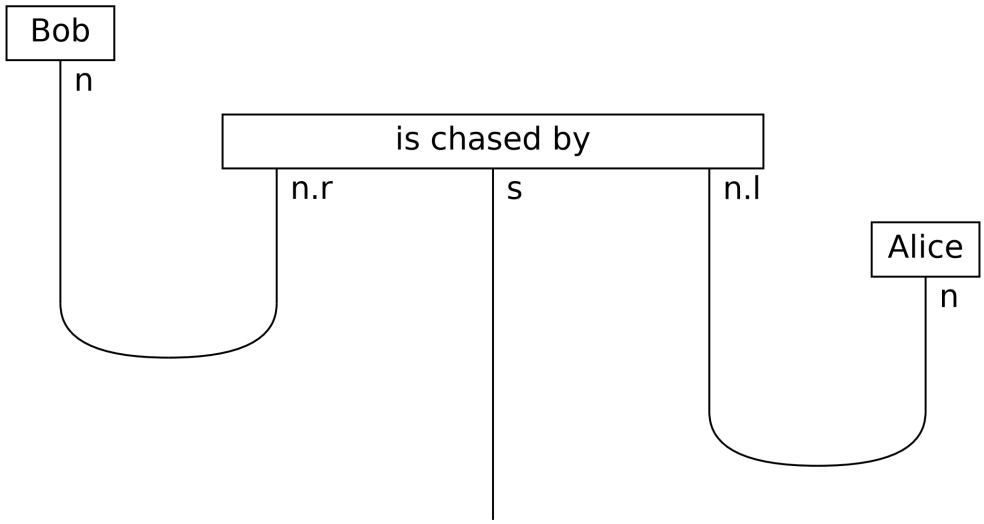


Figure 11: Passive example with Bob in the subject slot and Alice in a supplied oblique slot. “By Alice” is an English oblique phrase; INS is a modeling label in some illustrations, not an English instrumental inflection. The generic noun type does not enforce these role labels.

8 Compositional Distributional Semantics

A typed grammatical derivation can guide a semantic calculation. In the DisCoCat construction, a suitable interpretation maps grammatical types to semantic spaces and the derivation to a composition map [Coecke et al., 2010]. This separates the choice of a grammar from the choice of lexical representations.

8.1 An explicit tensor contraction

Let a_i and b_k represent the two noun vectors and V_{ijk} a transitive verb tensor, with the middle index ranging over sentence features. The example calculation is

$$m_j = \sum_{i,k} a_i V_{ijk} b_k. \tag{4}$$

The source helper `create_tensor_semantics()` constructs deterministic example tensors and evaluates their contraction. These are generated representations, not pretrained embeddings or estimates from a corpus. Exchanging the arguments changes the result when the selected tensor is sensitive to that exchange; the formalism does not guarantee every tensor will distinguish them.

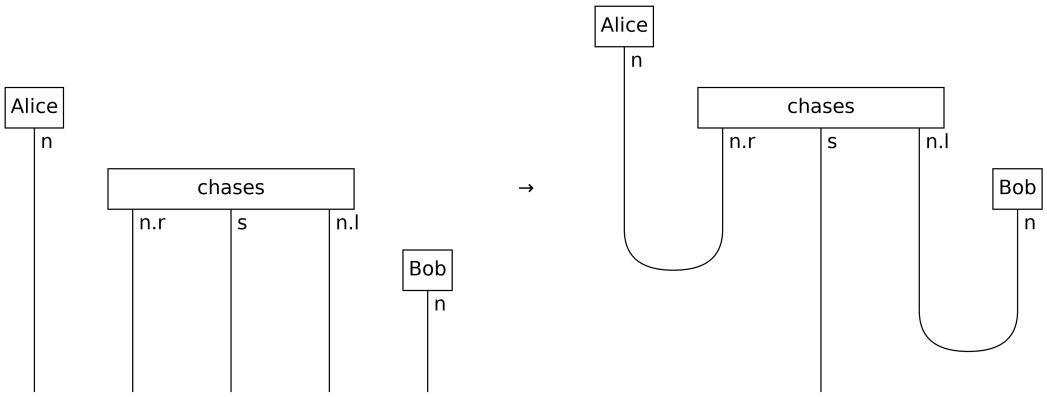


Figure 12: Word tensor product followed by grammatical contraction in the transitive example. The diagram shows how assigned types determine the slots used by semantic evaluation.

The diagrams make dependency on argument position inspectable. This is a representational advantage, not a measured interpretability result. Tensor entries can still be opaque, high-dimensional, or poorly estimated. Similarly, transformer attention is not automatically a functor or an enriched hom-matrix; such an identification needs domains, codomains, and law-preserving maps. A useful empirical comparison would train alternative composition models on the same data and test argument reversal and generalization on held-out constructions.

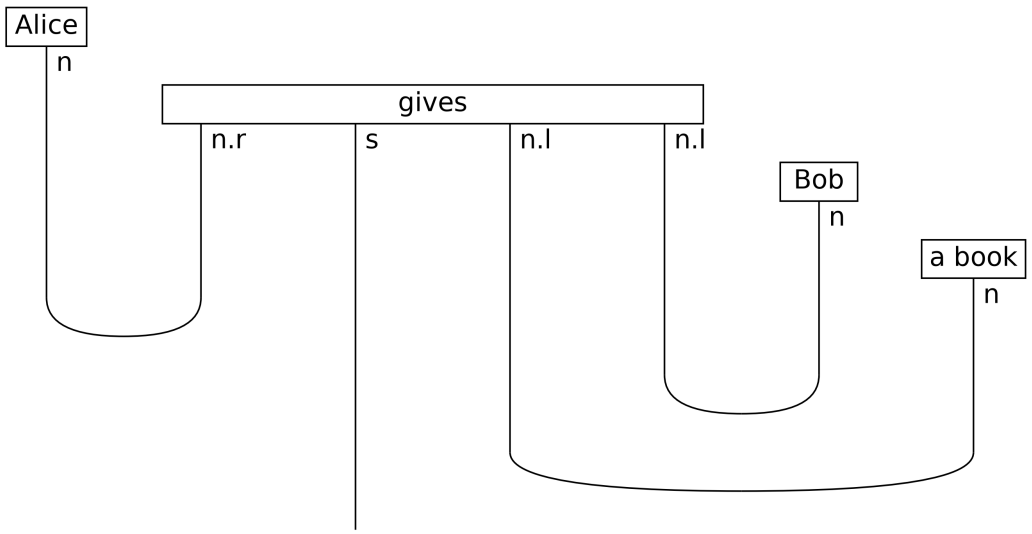


Figure 13: A ditransitive example with three explicitly supplied argument slots. Recipient and theme labels are metadata unless refined noun objects are introduced.

9 Duality Identities and Diagram Complexity

Cups and caps satisfy snake identities in the appropriate category with duals. A wire bent through a matching evaluation and coevaluation is equal to its identity morphism. The project constructs both sides and checks normal forms with DisCoPy; this is an executable algebraic example, not a new theorem.

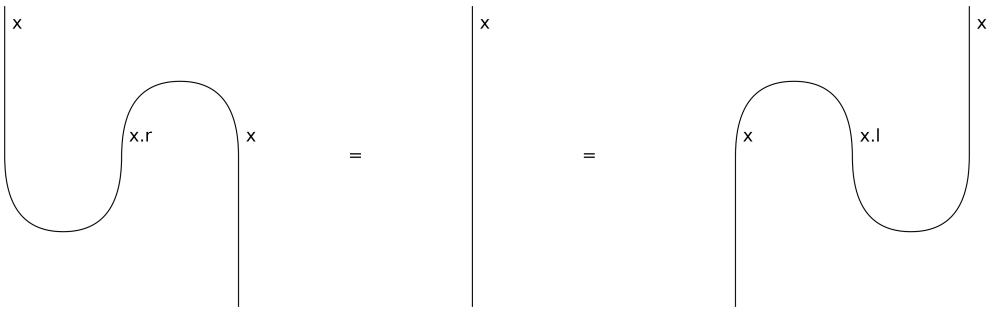


Figure 14: Executable snake-equation diagrams and the identity wire. Equality is tested within the selected DisCoPy calculus.

The complexity utilities report box count, cup count, cap count, depth, width, and normal-form size. The configurable score used by the examples is

$$\kappa(D) = 1N_{\text{lex}} + 0.5N_{\text{cup}} + 0.25N_{\text{cap}} + 0.1\, d(D). \tag{5}$$

Here `count_words` counts every non-Cup/non-Cap box; in diagrams containing a Swap, this is broader than a linguistic word count. The four coefficients are the declared defaults of `syntactic_complexity_score()` in the source, injected at build time; they are chosen weights, not a fitted psycholinguistic model. DisCoPy depth and width describe a particular representation and do not directly determine qubit count, hardware depth, processing time, or cognitive difficulty without an explicit compilation or measurement model.

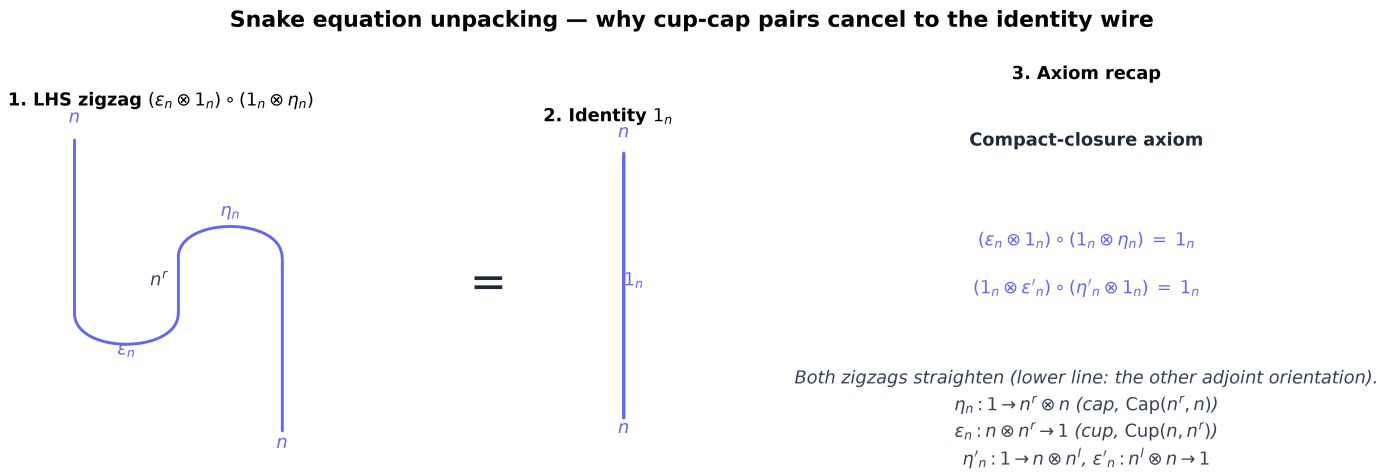


Figure 15: Stepwise illustration of a snake identity. The allowed categorical equation licenses straightening the wire; geometric resemblance alone would not.

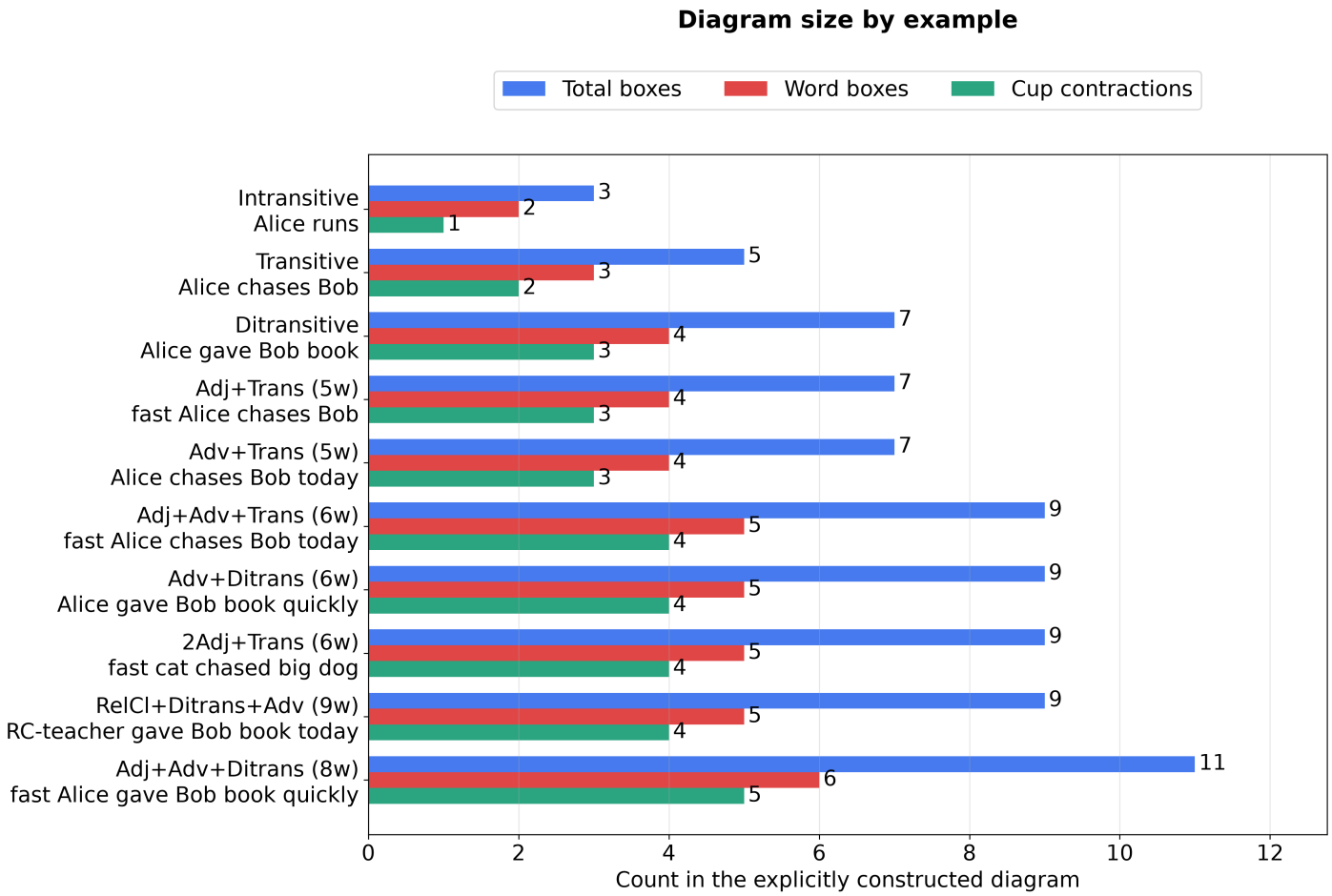


Figure 16: Complexity measurements for the generator's selected diagrams. Scores depend on the assigned lexical decomposition and the injected weights. The collection is illustrative rather than a benchmark sampled from a language.

The legacy `compute_pqc_decoherence_proxy()` derives a synthetic decoherence-rate field from a caller-supplied noise baseline, excess cups over caps, and an unsourced exponential amplification factor; the container also carries this section's base complexity score as its base field. Despite legacy field names, it computes neither topological homology nor a physical decoherence rate. No physical or security conclusion in this manuscript relies on that score.

10 Discourse: Entity Bookkeeping and State-Update Proposals

The same participant can occupy different grammatical roles across sentences. The `Discourse` helper records explicitly supplied entity names and role histories. For "Alice chases Bob. Bob fears Alice. Alice smiles," Alice's stored sequence is NOM, ACC, NOM. This is bookkeeping over supplied sentence objects. It does not infer that the pronoun "she" refers to Alice.

```
from src.diagrams.string_diagram import Sentence, Discourse
discourse = Discourse()
discourse.add_sentence(Sentence.transitive("Alice", "chases", "Bob"))
discourse.add_sentence(Sentence.transitive("Bob", "fears", "Alice"))
discourse.add_sentence(Sentence.intransitive("Alice", "smiles"))
assert [r.name for r in discourse.role_history["Alice"]] == ["NOM", "ACC", "NOM"]
```

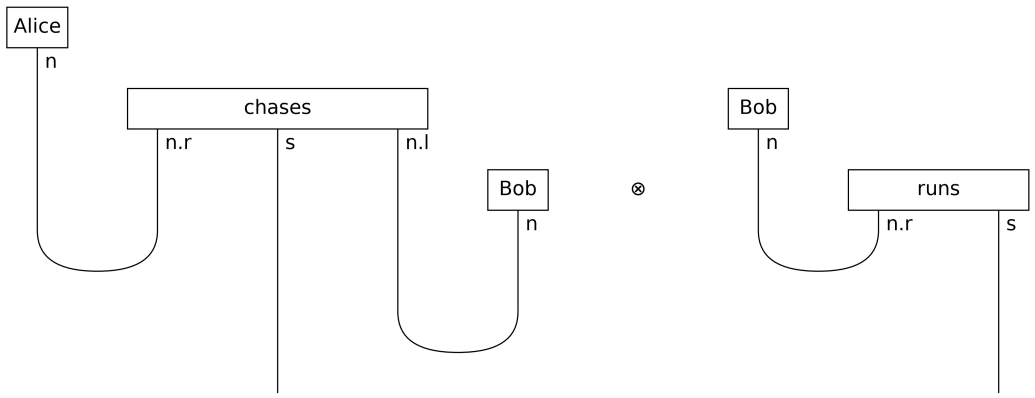


Figure 17: Tensor juxtaposition of two independently typed sentence diagrams. The output $s \otimes s$ does not itself encode coreference or a shared semantic memory.

A full discourse model would specify persistent semantic states and the update maps applied by sentences. It would also need a policy for aliases, repeated names, pronouns, and distinct participants with identical surface forms. These are open requirements here. Quantum compilation and trainability would introduce further choices of encoding, ansatz, observable, optimization, and hardware. Local observables alone do not justify an unconditional claim that barren plateaus are absent. No quantum circuit is trained in this project.

A related commitment appears at corpus scale in recent work on causal discourse extraction. Democritus groups paraphrase families of extracted causal claims into homotopy-localized classes, so that restated variants of one claim do not multiply the evidence count [Mahadevan, 2026b]. The present chapter's entity bookkeeping makes an analogous commitment at the opposite scale: role histories are recorded only for explicitly supplied names, sentence by sentence, and repeated spellings do not by themselves assert coreference. In both cases surface variation is treated as a modeling input rather than as new evidence — here by keeping each entity's history explicit, there by aggregating paraphrase families into localized claim classes. The shared boundary is also the same: neither layer claims semantic correctness of its identifications without an independent annotation layer, and neither supplies one.

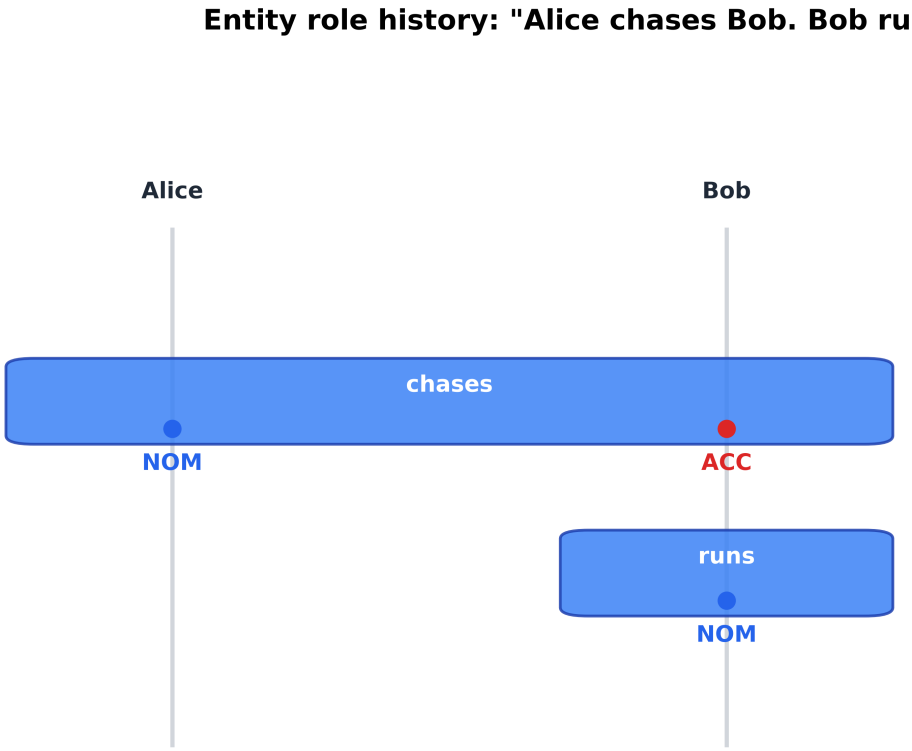


Figure 18: Native entity-history drawing for the two-sentence example. Persistent lines visualize explicit name matching in the supplied data.

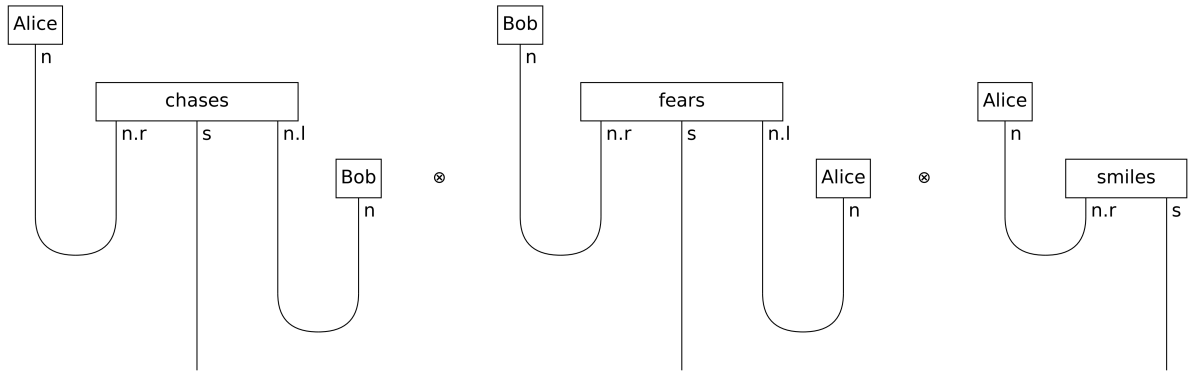


Figure 19: Three independently constructed sentence diagrams with repeated entity labels. Shared spelling is visible, but a tensor of sentence outputs does not implement a DisCoCirc state-update circuit.

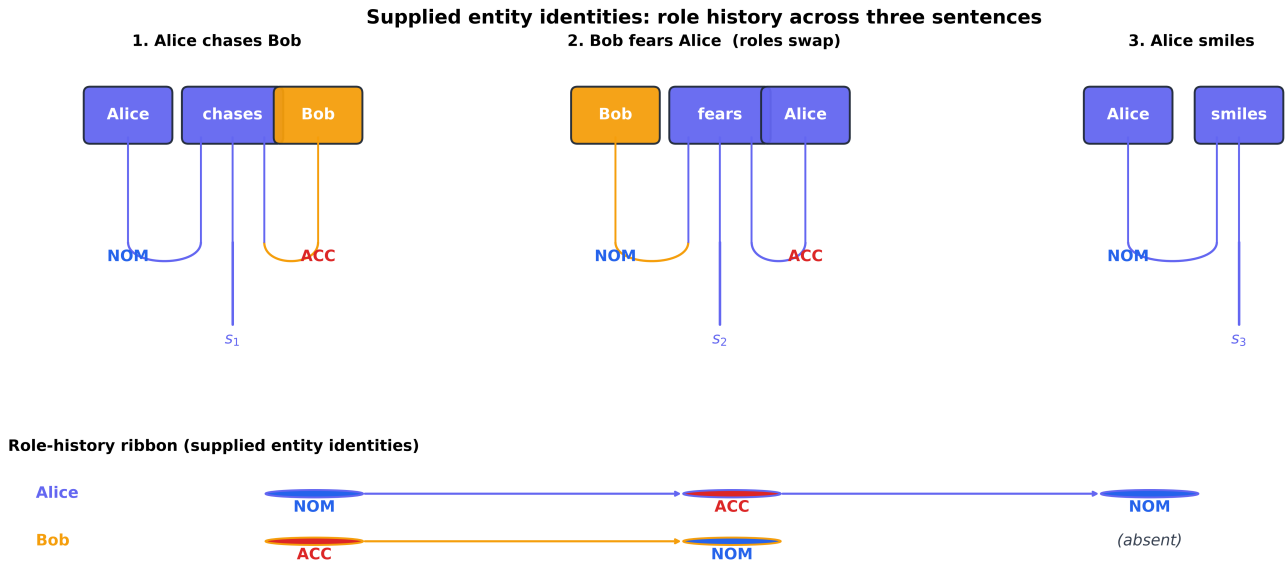


Figure 20: A separate history ribbon makes the supplied role changes readable. The construction illustrates entity persistence without claiming learned coreference or implemented Frobenius updates.

11 Candidate Similarities and $[0, 1]$-Enrichment

For enrichment over the ordered monoid $([0, 1], \cdot, 1)$, a finite hom-matrix Z must satisfy

$$Z_{ii} = 1, \qquad Z_{ik} \geq Z_{ij} Z_{jk} \quad \text{for every } i, j, k. \tag{6}$$

The inequality follows the order convention used here. Symmetry is optional. Normalized conditional probabilities, cosine similarities, and attention weights do not generally satisfy these axioms. They are candidate inputs requiring separate justification.

The standard 8-role matrix in `src/enriched_cat/enriched.py` is hand-selected and fails some composition inequalities. For example, its NOM-to-DAT value is 0.45, whereas the NOM-to-ACC-to-DAT product is $0.85 \times 0.55 = 0.4675$. The failure is retained as a useful counterexample; the matrix is not described as an English estimate or a valid enriched category merely because its entries lie in $[0, 1]$.

`composition_closure()` computes the maximum path product using a Floyd–Warshall update $Z_{ik} \leftarrow \max(Z_{ik}, Z_{ij} Z_{jk})$. It returns a new matrix and leaves the raw input intact. With entries at most one and unit diagonals, cycles cannot improve a path product beyond a cycle-free representative. The resulting finite closure satisfies the multiplicative path inequalities up to the stated numerical tolerance.

```
from src.enriched_cat.enriched import standard_enriched_category
candidate = standard_enriched_category()
closed = candidate.composition_closure()
assert not closed.full_composition_check()["violations"]
```

Closure is a mathematical operation, not statistical validation. It raises selected similarities and can alter their interpretation, spectrum, magnitude, and clusters. Both raw and closed matrices should be retained in empirical work. Bradley, Terilla, and Vlassopoulos construct enriched structure from suitable text-extension probabilities [Bradley et al., 2021]; this does not identify the present role matrix with their construction.

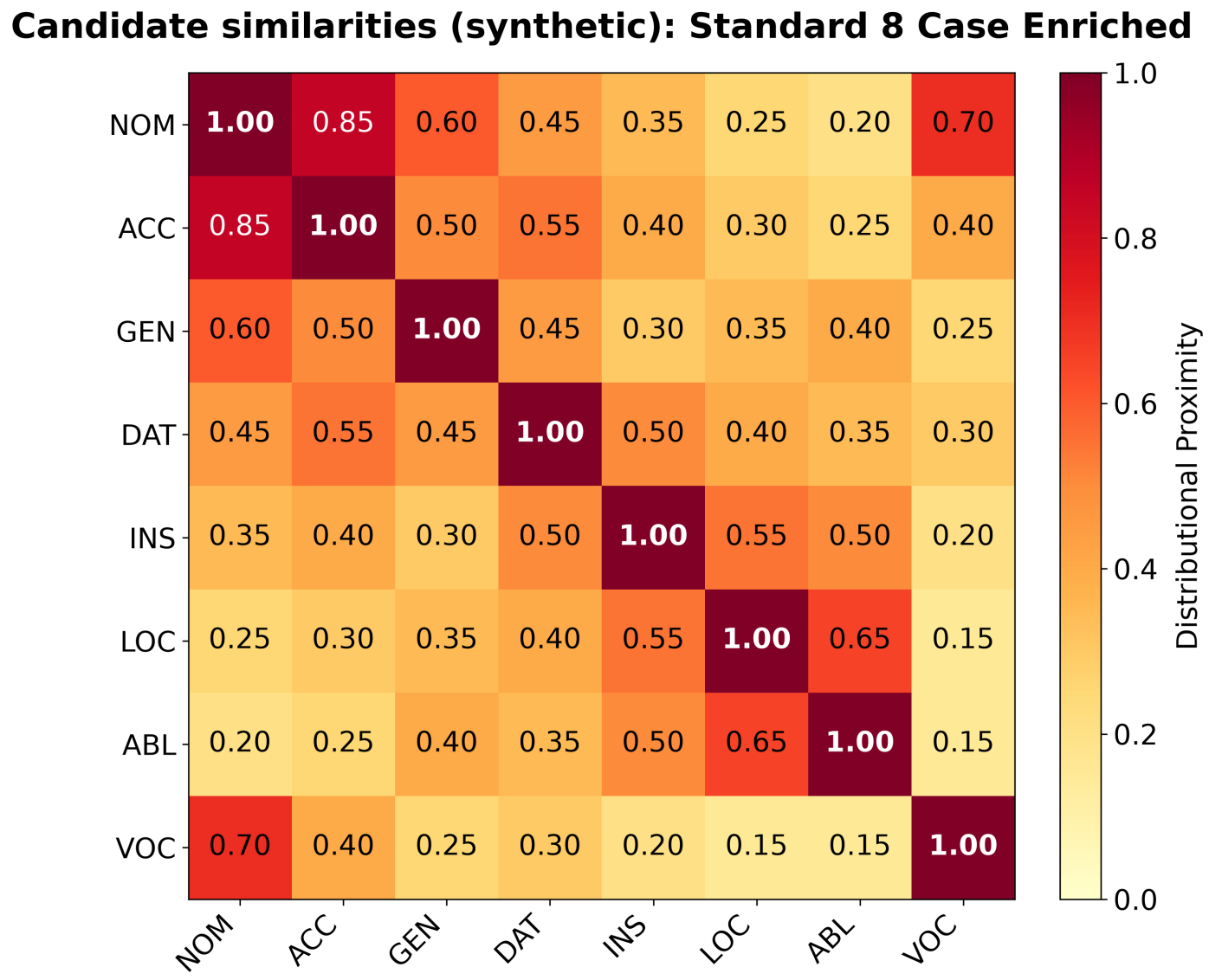


Figure 21: Synthetic candidate similarity matrix. Read entries as selected numerical examples. The raw matrix fails composition closure; neither its colors nor its linguistic labels provide empirical provenance.

12 Magnitude, Weightings, and Numerical Sensitivity

For a finite similarity matrix Z, a weighting w and coweighting v satisfy $Zw = \mathbf{1}$ and $v^T Z = \mathbf{1}^T$. When both exist, their sums agree because $v^T Z w$ can be evaluated in either order. If Z is invertible, the matrix magnitude is

$$|Z| = \mathbf{1}^T Z^{-1} \mathbf{1}. \tag{7}$$

For a two-role example $Z = \left(\begin{smallmatrix} 1 & a \\ a & 1 \end{smallmatrix}\right)$ with $0 \leq a < 1$, the weighting is $(1/(1+a), 1/(1+a))$ and the magnitude is $2/(1+a)$. At $a = 1$, the inverse does not exist, yet weightings and coweightings do exist and have sum one. Singularity therefore does not by itself make magnitude undefined.

The implementation uses an inverse for well-conditioned nonsingular matrices and a pseudoinverse when required. In the latter case it checks the left and right weighting residuals; a least-squares vector that fails these equations is not accepted as magnitude. Matrix mutation invalidates the cached inverse. A near-singular result remains sensitive to perturbations, and residual tolerances are numerical criteria rather than exact proofs.

For the raw illustrative matrix, the generated inverse-sum statistic is 2.49608, with two-norm condition number approximately 91.8505. Its object count minus that statistic is 5.50392. These describe this chosen matrix; the raw matrix's failed enrichment axioms preclude presenting the result as a categorical invariant of a validated linguistic model.

Magnitude homology is a distinct graded construction [Leinster and Shulman, 2021]. The project does not construct its chain complex, differentials, or homology groups. A cup-minus-cap count is not a substitute. Likewise, a magnitude deficit is not automatically an entropy, a redundancy estimate, a reanalysis cost, or a security margin. Those interpretations require additional assumptions and evidence.

For positive hom-values, the transformation $d(i, j) = -\log Z_{ij}$ converts the multiplicative composition inequality to a triangle inequality. Zero hom-values correspond to infinite distance, and symmetry is still an additional condition. This connection explains a mathematical relationship between similarity and distance; it does not identify arbitrary attention matrices with metric spaces.

13 Topos-Theoretic Bridges: Requirements and Present Scope

A geometric theory has a specified signature and axioms in geometric logic. Its classifying topos represents its models across suitable toposes [Caramello, 2018, sec. 2.1.2]. Morita equivalence concerns equivalence of classifying toposes, or equivalently the appropriate natural equivalence of model categories [Caramello, 2018, sec. 2.2.2]. Caramello's bridge programme uses alternative presentations of a common topos to transport invariant information [Caramello, 2016].

This is stronger than similar vocabulary or matching counts [Caramello, 2018, sec. 2.1.7, sec. 6.1.1, sec. 7.1, sec. 10.1]. Adding a definable relation symbol can change signature size without changing the modeled structure. Conversely, two theories can have equal numbers and arities of symbols while imposing incompatible axioms. Counts of sorts, relations, or axioms are therefore neither necessary nor sufficient conditions for Morita equivalence.

13.1 What the software represents

`GeometricTheory` stores names, relation arities, and textual axiom descriptions. `ClassifyingTopos` is a compatibility name for a presentation-profile container; it does not construct a Grothendieck topology, a category of sheaves, or a classifying universal model. The standard builder supplies 8 sorts and 8 relations; the minimal builder supplies 3 sorts and 3 relations. These are presentation statistics only.

`compare_theory_presentations()` compares those statistics. The deprecated `check_morita_equivalence()` retains the same profile-comparison result with a warning; its Boolean must not be interpreted as Morita equivalence. `bridge_transfer()` reports `morita_equivalent=None` and `transfer_possible=False` because no equivalence witness is supplied, even when profiles match.

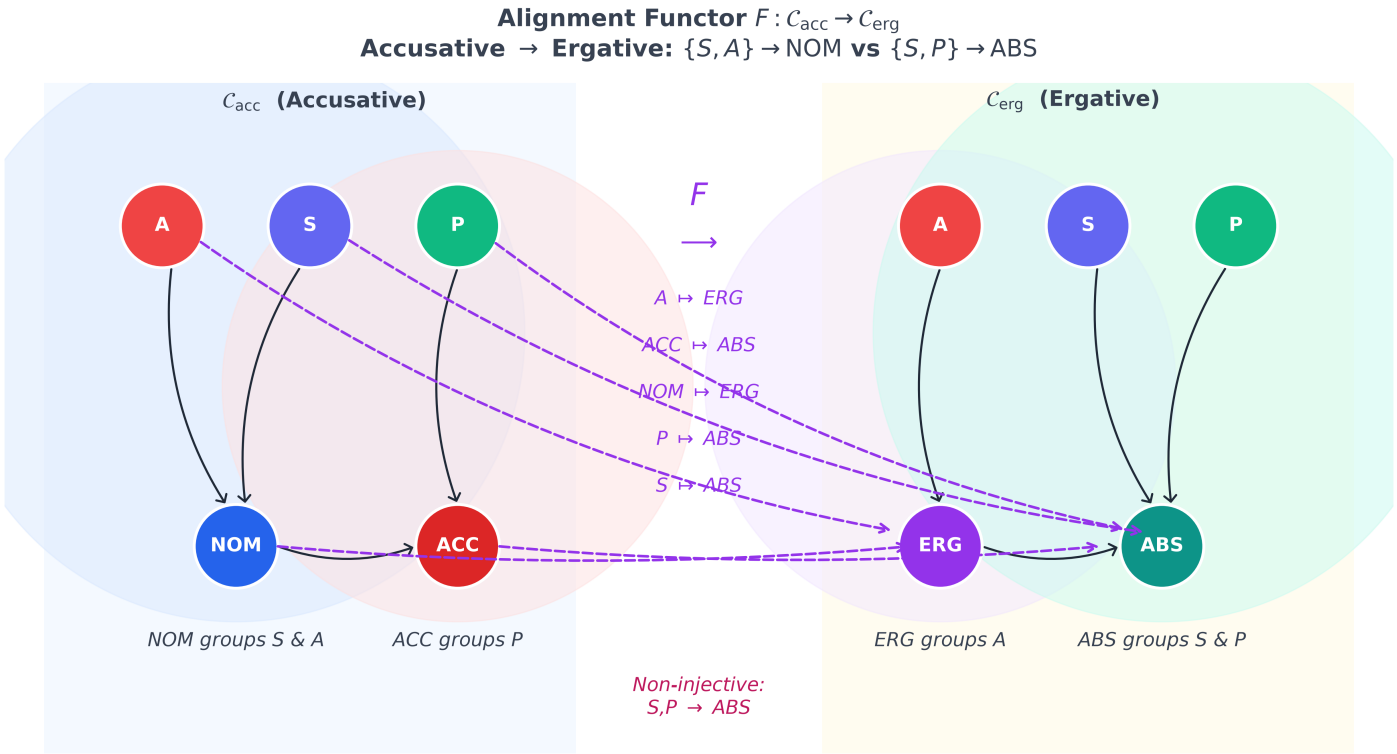


Figure 22: Schematic alignment mapping. A map between selected role labels illustrates a candidate correspondence; it does not construct a geometric morphism or a topos-theoretic bridge.

13.2 What a bridge would require

For two candidate case theories, a substantive result would specify their geometric syntax, exhibit the relevant models or sites, construct comparison functors, and prove the required equivalence and coherence. A transferred claim must then be identified as invariant under that equivalence. No such witness is constructed here. A two-object picture or a matching profile cannot replace these steps.

A complementary conditional-grounding pattern appears in the causal-discourse setting. Democritus states a relative grounding functor from localized discourse to observational-equivalence classes of causal models as a conditional propo-

sition, with its hypotheses — cocompleteness and density conditions among them — left explicitly to check rather than asserted [Mahadevan, 2026b]; the work forms part of a broader categorical-AI research program developed in dated draft editions [Mahadevan, 2026a]. This matches the present chapter's posture. `bridge_transfer()` likewise reports an unknown equivalence and performs no transfer: in both cases the topos and bridge language functions as a specification of what evidence a real grounding or transfer would require, with the missing evidence named rather than assumed.

Topos logic also needs careful separation from probability and quantum mechanics. Heyting algebras are distributive, although excluded middle need not hold. Intuitionistic logic is not made classical by "sheaf decoherence," and no cohomological collapse or thermodynamic limit follows from the present code. Formalizing linguistic theories and testing which invariants survive is a research direction, not an accomplished unification.

14 Bayesian Case-Role Beliefs and Active-Inference Motivation

Active inference supplies a motivation for relating generative models, uncertainty, and action [Friston et al., 2017]. The implemented cognitive core is narrower: finite categorical beliefs, Bayesian likelihood updates, KL divergence, variational free-energy evaluation, and caller-defined policy scores.

For prior probabilities p_i and nonnegative likelihoods L_i with positive evidence $e = \sum_i p_i L_i$, the update is $q_i = p_i L_i / e$. Likelihoods need not sum to one over states. A negative likelihood is invalid even where the prior is zero, and an all-zero or disjoint-support likelihood is an error rather than a reason to invent a uniform posterior.

For a fixed generative pair (p, L),

$$F(q) = D_{\mathrm{KL}}(q \| p) - \sum_i q_i \log L_i = D_{\mathrm{KL}}(q \| p(\cdot \mid o)) - \log e. \tag{8}$$

The identity holds with consistent support conventions. It establishes the usual bound for this fixed model; it does not imply that free energies computed under different incoming priors and observations must decrease. Zero-probability terms are masked rather than evaluated as $0 \log 0$.

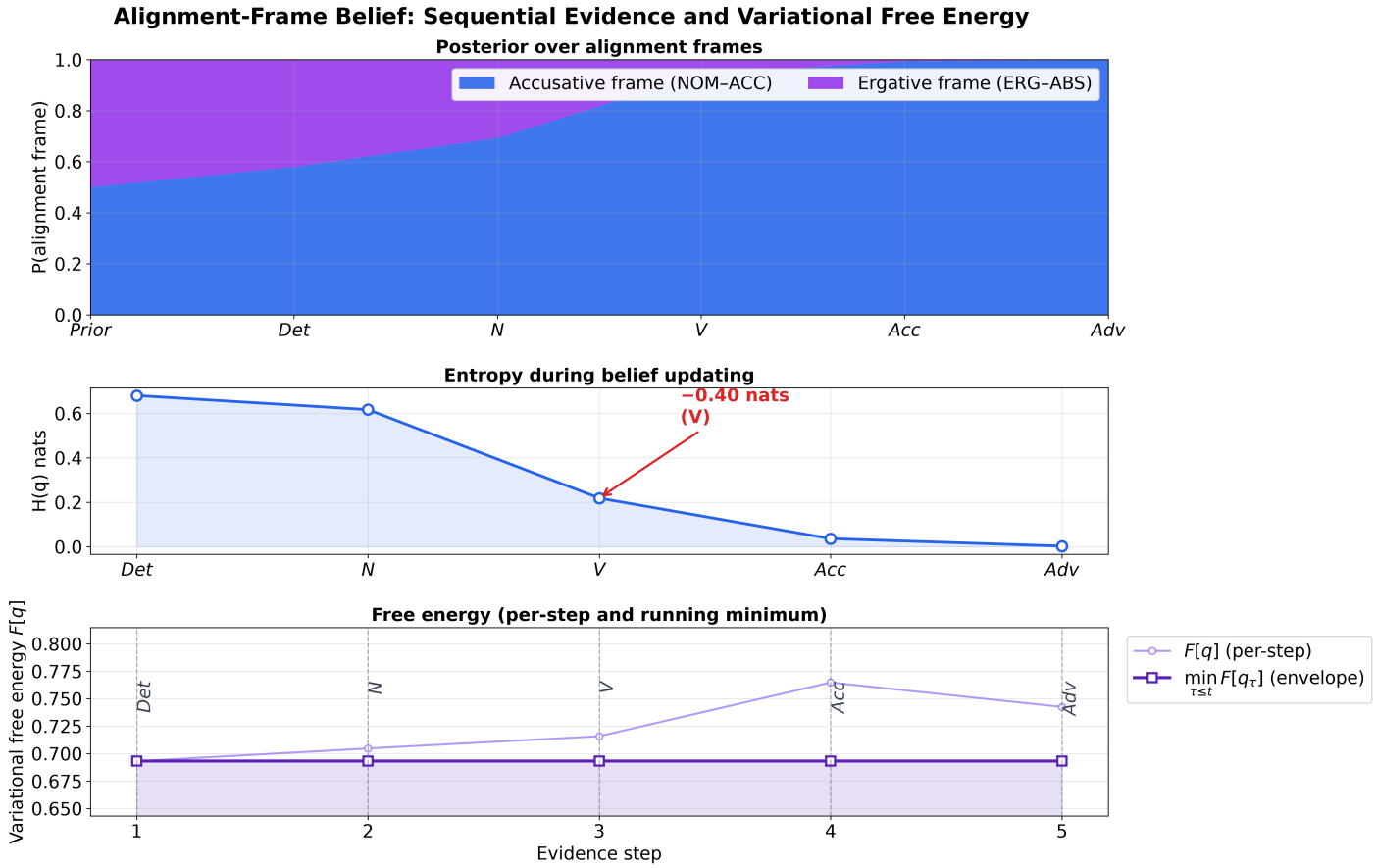


Figure 23: Synthetic filtering over candidate alignment-frame labels. Likelihood vectors are hand-selected; the plot demonstrates the supplied update rule rather than an empirically inferred change in a language's alignment.

`sequential_belief_update()` consumes each supplied likelihood once. `distributional_case_assignment()` repeatedly consumes the same likelihood, optionally with a Markov prediction step; its iteration parameter is therefore an evidence-repetition choice, not just an optimizer budget. A single categorical belief indexes the candidates declared by the caller. It cannot simultaneously stand for every noun's case, the whole parse tree, and an alignment system without an explicit joint model.

Policy selection utilities evaluate supplied surprise, information-value, and utility vectors. A full active-inference model would need a specified generative process, observation predictions under policies, and preferences. Those components are not inferred from the graph drawing.

15 Cognitive Hypotheses and Testable Comparisons

A diagram can make a relation easy to inspect when its visual arrangement exposes a needed constraint. This motivates a hypothesis about case diagrams, not a universal claim that the brain prefers them. Drawing readability, algebraic correctness, and human reasoning performance are different outcomes.

A controlled study could compare matched diagrammatic and textual explanations of the same small role structures. Accuracy, response time, and transfer to new examples should be measured separately, with training time, diagram familiarity, language background, and visual complexity controlled. Counterexamples in which a diagram hides a needed distinction are as informative as successes.

The implementation's prediction-error proxies provide another source of hypotheses. Multiplying a mismatch by a selected weight demonstrates sensitivity to that weight by construction. It does not establish that the weight is neural precision or that the output is an N400 or P600 amplitude. The scalar example is

$$\mathrm{PE} = w\,|\mu_{\mathrm{pred}} - \mu_{\mathrm{obs}}|. \tag{9}$$

To evaluate a physiological interpretation, a study would need a dataset, preprocessing specification, electrode and time-window definitions, parameter estimation on training data, and held-out comparison against simpler alternatives. The current project includes none of those measurements. Consequently the ERP-inspired functions in sec. 16.5 return uncalibrated model quantities.

The software suite exercises specific algebraic and numerical contracts. It does not certify every statement in this review, independently verify the implementation, or establish cognitive privilege. The reproducibility appendix sec. 25 states what the tests cover and what empirical claims remain outside their scope.

16 Distributional Utilities and Synthetic Filtering

Distributional active inference is an existing research programme [Akgül et al., 2026]. The local `src/daif/` package is a collection of experimental utilities inspired by distributional representations; it is not a reproduction of that paper's complete algorithm. Several compatibility names predate the present clarification. This section defines the quantities actually computed.

16.1 Finite score distributions and projection

For role probabilities q, a row-stochastic transition matrix T, dimensionless score vector R, and $\gamma \in [0, 1]$, `push_forward_return()` forms

$$z = R + \gamma T^T q, \qquad \mu = \sum_i q_i \delta_{z_i}. \tag{10}$$

The mass at z_i is q_i. Mean and variance use exact finite weighted sums. The stored quantile values use the generalized inverse of this discrete CDF, removing zero-mass support. They do not interpolate new values between distinct atoms. As a verified numeric instance of this generalized inverse, for equal mass at the support endpoints 0 and 10, the midpoint levels 0.125, 0.375, 0.625, 0.875 yield 0, 0, 10, 10.

Despite its legacy name, `distributional_bellman_operator()` propagates beliefs and repeats this score calculation. It does not back up state-conditioned future-return distributions or sum discounted rewards. In the canonical counterexample configuration — uniform transition entry 0.5, uniform two-state belief, reward entries 1 and 0, and discount 0.9 — its score mean is 0.95. The true infinite-horizon expected return for that Markov reward process, computed from $(I - \gamma T)^{-1} R$, is 5. This counterexample rules out treating the utility as a Bellman solver; both quoted numbers are recomputed from the same declared configuration at build time.

`categorical_return_distribution()` and `DistributionalReturn.to_categorical()` share a C51-style projection: clip supplied samples to fixed support, then split each sample's mass between adjacent atoms. Values outside the support contribute to endpoint atoms rather than disappearing. Stored quantiles do not reconstruct every property of the original law; projection of equally weighted quantile samples is an explicit approximation.

16.2 Pairwise quantile regression and distortion

For current values θ_i, target samples y_j, and midpoint levels τ_i, the update uses every pair $\delta_{ij} = y_j - \theta_i$, as motivated by quantile regression [Dabney et al., 2018b]. Define H_κ as the ordinary Huber loss. The implemented coordinate update is

$$\theta_i' = \theta_i + \alpha \frac{1}{M} \sum_j |\tau_i - \mathbf{1}_{\delta_{ij} < 0}| \ \mathrm{clip}(\delta_{ij}, -\kappa, \kappa). \tag{11}$$

Here the Huber loss is unnormalized: its derivative is clipped at κ. The learning-rate convention absorbs averaging over current coordinates. Comparing with a paper or implementation using H_κ/κ requires corresponding rescaling of α. The output is sorted to retain quantile order. Current and target sample counts can differ.

`implicit_quantile_network_update()` applies pairwise Huber updates with distorted current probability levels; it contains no neural network. Target values are equally weighted samples; supplied target levels are validated metadata, not quadrature weights. The IQN literature concerns learning quantile functions [Dabney et al., 2018a]. In this helper, for $0 < \eta < 1$, optimistic distortion $1 - (1 - \tau)^{1/\eta}$ emphasizes upper values, while pessimistic distortion $\tau^{1/\eta}$ emphasizes lower values. Both reduce to the identity at $\eta = 1$; the direction reverses if $\eta > 1$, so the mode name alone is insufficient without its parameter.

16.3 Filtering, single-factor messages, and factor scores

At each filter step the code computes $p^- = T^T q$ and $q' = L \odot p^- / \sum_i L_i p_i^-$. It stores the updated belief before any stopping decision. Diagnostics include the complete posterior trajectory, the number of updates, and `stop_reason`. Stopping requires small changes in both belief and free energy; an increase in free energy alone is not an error because successive priors change.

`variational_message_passing()` is a single-factor calculation with an implicit uniform prior:

$$q_i = \mathrm{softmax}_i(\lambda_{\mathrm{lik},i} o_i). \tag{12}$$

It also returns $\lambda_{\text{prior}} + \lambda_{\text{lik}}$ as bookkeeping. Prior precision does not determine a nonuniform categorical prior. Because the incoming message is constant, extra sweeps do not perform general loopy inference.

The legacy `bethe_free_energy()` is now also exposed as `factor_consistency_score()`. Its actual formula is

$$C(q) = \sum_a D_{\text{KL}}(q \| \hat{f}_a) - \left(\sum_i (d_i - 1) \right) H(q). \tag{13}$$

There is only one categorical distribution q, and $\hat{f}_a$ is a normalized supplied vector. This is not a Bethe free energy: no joint factor marginals, individual variable marginals, or factor potentials are represented. Adjacency must be binary, and negative factor weights are rejected.

`expected_information_gain()` returns each observation's contribution $p(o) D_{\text{KL}}(q(\cdot \mid o) \| q)$. Its sum is mutual information only if the supplied likelihood columns sum to one across a complete observation alphabet. An arbitrary list of candidate likelihood vectors is not automatically such an alphabet.

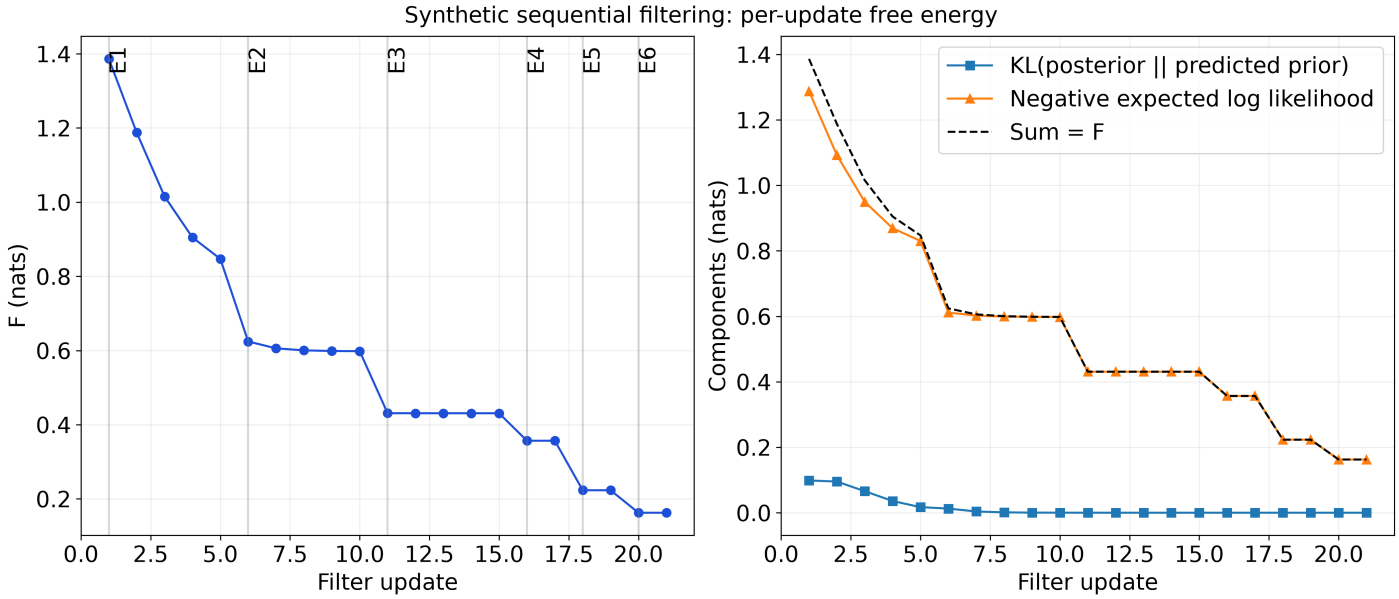


Figure 24: Computed per-update free energy and its KL/data-fit decomposition for synthetic repeated-observation filtering. Each evidence vector is used for up to 3 updates before the next vector. There is no fitted convergence envelope or fixed-objective convergence claim.

16.4 Caller-defined policy scores

`G_policy()` evaluates $-q \cdot \ell - q \cdot e - \gamma q \cdot u + \beta \operatorname{Var}(Z)$ for supplied finite vectors ℓ, e, u, with Z the scalar score distribution of the chosen policy. This can express a chosen surprise, information-value, utility, and risk tradeoff. It is not a general expected-free-energy derivation. Expected log likelihood of one observation is not ambiguity entropy, and posterior entropy is not information gain. Units must be made compatible by the coefficients.

Policy probabilities are computed by a stable softmax of negative scores. Subtraction precedes temperature scaling to avoid overflow. Low temperature concentrates mass on minima, including equal treatment of tied minima; no optimality theorem about the environment follows.

16.5 Prediction-error and ERP-inspired proxies

The scalar error is the distributional prediction error (DPE), weighted surprisal $-w \log q_i$, with a small positive probability floor in the compatibility API. A separate function uses wW_1 between two reconstructed quantile distributions. These are distinct mismatches, not interchangeable derivations of one physiological observable.

The N400-inspired proxy is $-s\lambda|m - m_0|$. The P600-inspired proxy is $sk \max(0, \lambda_{\text{post}} - \lambda_{\text{prior}})$DPE. Severity s, scaling k, and precision values are caller inputs. The waveform utility places these amplitudes in Gaussian templates with specified

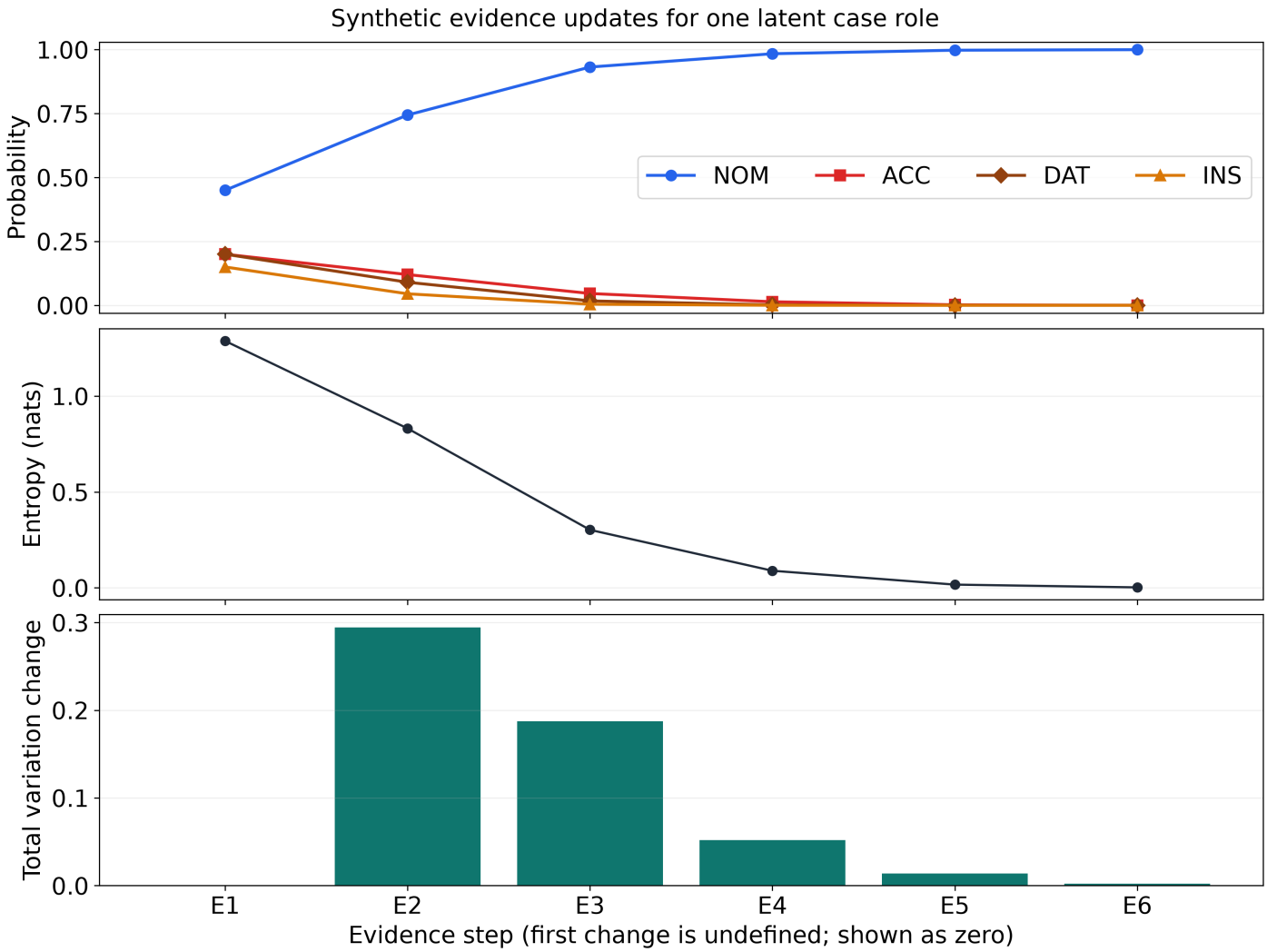


Figure 25: A separate sequence consumes 6 hand-selected likelihood vectors once each. Panels show probabilities, entropy, and total variation between adjacent posteriors for one latent role. The first displayed change is a zero placeholder; it is not a measured change from an omitted prior. No percentile bands or token-specific parses are inferred.

millisecond centers and widths. Millisecond timing labels do not calibrate amplitude to microvolts; compatibility fields ending in `_uV` still contain uncalibrated values.

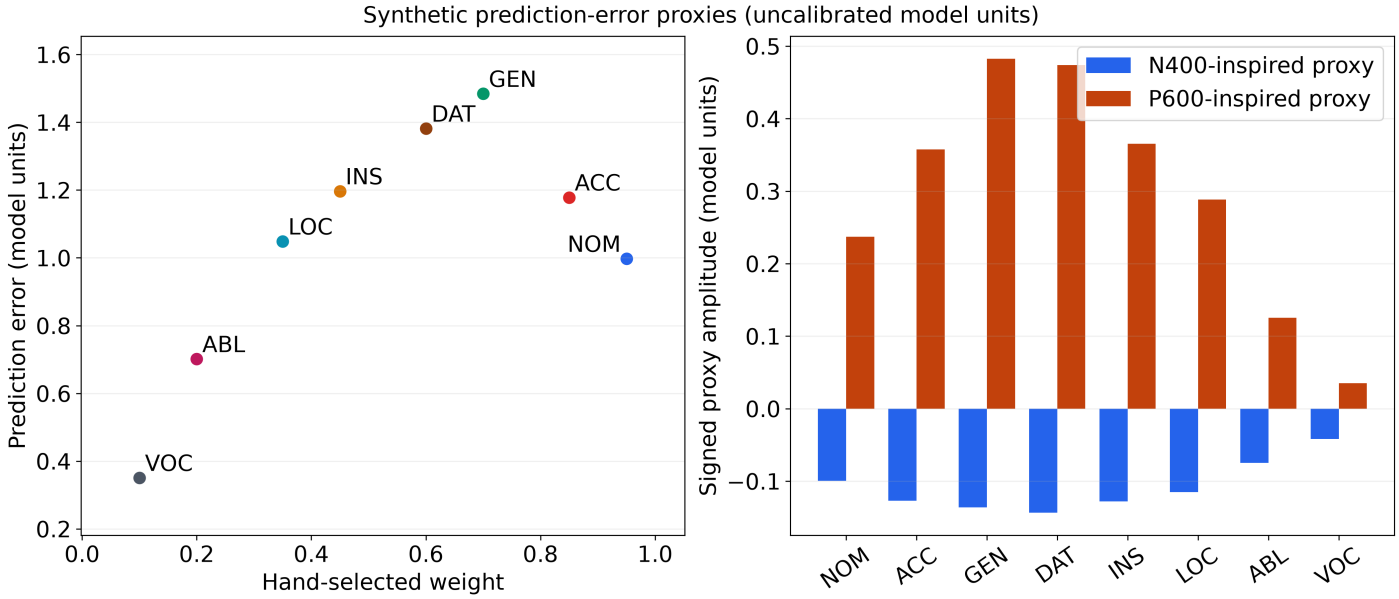


Figure 26: Synthetic prediction-error and amplitude proxies across role labels. Hand-selected weights, beliefs, and severity values determine the outputs. Amplitudes are model units; no EEG dataset, empirical error bars, or comparison with literature microvolt ranges is shown.

16.6 Metrics and reconstruction conventions

Wasserstein distance uses both stored probability-level grids, even when their lengths match. Between levels it linearly interpolates quantiles and holds tails constant to zero and one. Integration for orders one and two is exact for these reconstructed laws. It is not exact for an unknown originating distribution, and no universal convergence rate is asserted.

Histogram KL and entropy depend on the binning and smoothing parameters. `quantile_coverage()` compares observations with supplied quantiles; calling it does not itself create held-out calibration data. `convergence_diagnostics()` reports changes in a supplied series; a small change is not proof of a unique fixed point.

16.7 Scope and extensions

A genuine distributional-control experiment would need a reward process, state-conditioned return laws, a justified Bellman or inference operator, parameter learning, baseline comparisons, and independent evaluation data. An EEG experiment would additionally require physiological calibration. Neither is supplied here. The present utilities are useful as explicit numerical examples because these missing steps are visible. The seeded synthetic computations in sec. 17 characterize the filtering, calibration, projection, and sensitivity behavior of these utilities on their declared inputs; they remain synthetic and add no empirical evidence.

16.8 CEREBRUM as a design vocabulary

The 8 familiar case labels can organize a proposed agent architecture: actor, affected content, recipient, instrument, context, source, possession, and address. This analogy does not establish eight neural modules or a universal factorization of cognition. The software does not implement a complete CEREBRUM agent. Proposed architectural mappings should be evaluated as engineering choices with explicit interfaces and tasks.

17 Synthetic Statistical Behavior of the Utilities

The preceding sections define what each utility computes. This section reports what those utilities *do* on their declared synthetic inputs, computed by the seeded runner in `src/experiments/`. Every quantity here is synthetic: inputs are fixed or seeded constructions, replication is resampling over seeded runs, and intervals summarize replication-to-replication variation of the computation itself. No corpus, participant, physiological, hardware, or deployed-agent evidence is involved, and none of these numbers estimates a property of any language or brain.

The runner executes configured arms with a seeded PCG64 generator (seed 20260907, 256 replicates per study) and emits a schema-annotated result file whose configuration hash is `18c7278ee89606d88f5c0a147f63deeba17c692168006facf88cea182554c2f9`; the manuscript quotes only registry-declared identifiers from that file through injection. Replicate-level means carry two-sided 95% normal-approximation intervals (normal quantile 1.95996); point maxima and residuals are reported as descriptive bounds without intervals. There are no p-values, significance labels, or comparisons with empirical work in this section, and all configured arms are reported without multiplicity adjustment.

17.1 Filtering and transition ablation

The filtering study assimilates a known synthetic likelihood over 3 roles, repeating that same-likelihood assimilation 3 times per replicate with a tight convergence threshold, and compares the stored trajectory with the exact repeated-Bayes closed form. Mean posterior probability of the true role after the final assimilation is 0.997309 with interval [0.996974, 0.997644]. A paired arm re-runs the same repeated assimilation with a banded informative transition on identical draws, so the paired mean log-probability difference -0.0214211 nats with interval [-0.0225471, -0.0202951] isolates the effect of changing the transition while the likelihood is held fixed. The stored trajectory agrees with the closed form to a maximum deviation of 3.33067e-16, and the mean final free energy is 0.157986 nats with interval [0.150038, 0.165933]. Posterior mode match is reported descriptively as 1.

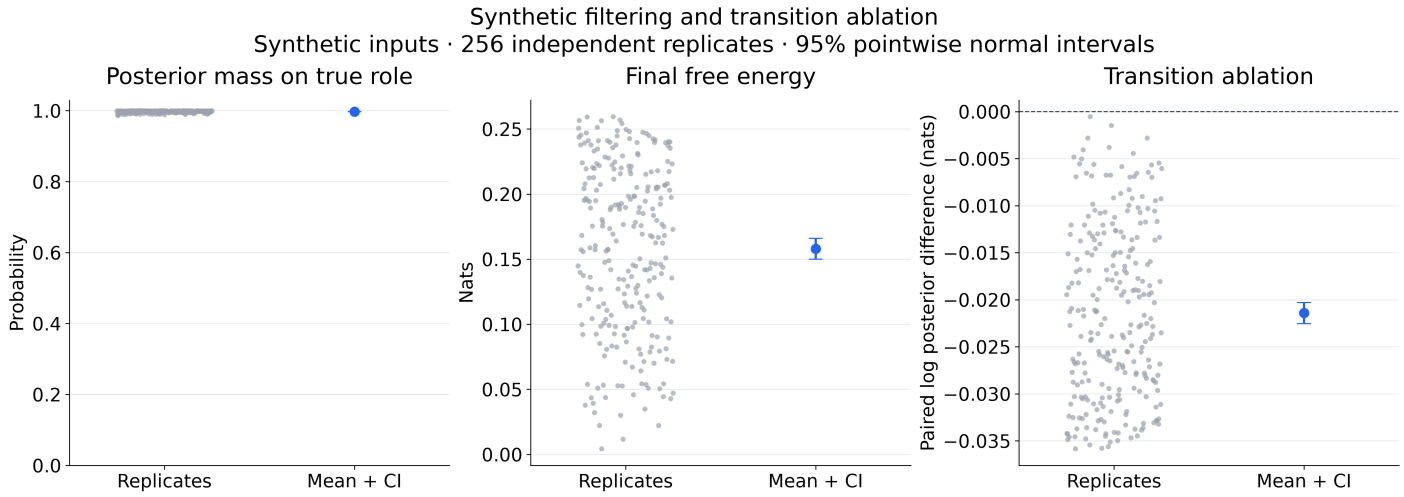


Figure 27: Synthetic Bayesian transition ablation across seeded replicates: identity-transition assimilation versus a banded informative transition on identical draws, with replicate-to-replicate uncertainty bands. The bands are replication intervals of a synthetic computation, not sampling error from empirical data, and the plotted quantities are model probabilities and nats.

17.2 Calibration of quantile coverage

For a finite discrete return law, the exact coverage target of a quantile level is its distribution function $F(Q(\tau))$, which need not equal τ under the discrete generalized inverse; that atom gap is a separate descriptive quantity. With 19 levels and 200 observations per replicate, observed coverage of the stored quantiles is compared with that exact target. The mean absolute deviation from the exact discrete-law target is 0.0202648 with interval [0.0187924, 0.0217371]; this deviation quantifies Monte Carlo sampling error of the empirical coverage estimate around the exact discrete target, not distance from a continuous coverage convention. The largest per-replicate deviation is 0.0698747, and the largest per-level gap between mean observed coverage and the mean exact target is 0.00392405. The mean exact atom gap $F(Q(\tau)) - \tau$ itself is 0.139219, and its largest value is 0.624411; both are deterministic discretization properties of the stored atoms and carry no sampling error. These diagnostics are properties of the stored representation and the seeded sampling scheme, not estimates from data.

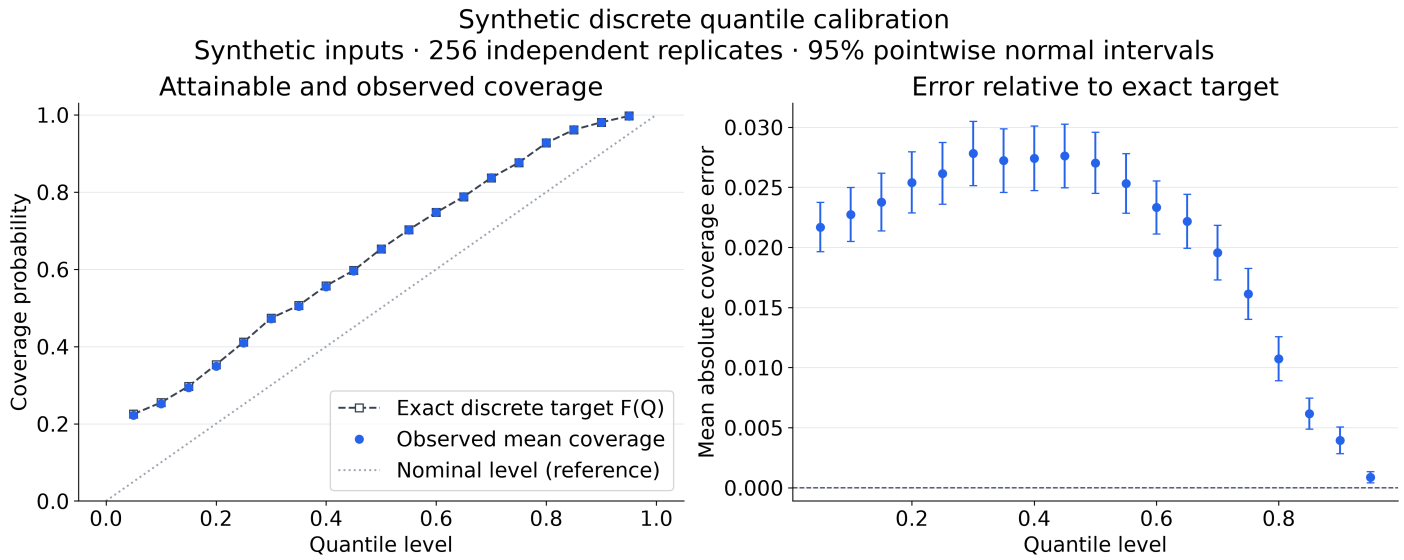


Figure 28: Empirical discrete coverage of stored quantile levels against the exact $F(Q(\tau))$ target of the same synthetic law, with paired per-replicate sampling errors. Coverage targets are exact arithmetic properties of the discrete law; no empirical calibration data is used.

17.3 Projection and quantile identities

The C51-style projection must conserve total mass and preserve the mean of clipped samples. Across seeded replicates and configured atom counts, the maximum mass residual is 2.22045e-16, the mean mean-exactness residual is 6.78033e-17 with interval [6.10892e-17, 7.45174e-17], and the largest such residual is 3.33067e-16. The Wasserstein reconstruction satisfies its self-distance identity to at most 0. The pairwise Huber update agrees with its finite-difference numerical gradient characterization to at most 8.15287e-11; this is a numerical agreement check of the implemented gradient, not a machine-precision identity proof. The remaining registered identifiers of this and the preceding subsections — the brute-force Wasserstein residuals, the projection clipping and refinement summaries, the quantile shift and triangle-inequality checks, and the per-replicate filtering summaries — are recorded in the machine-readable variable registry (output/experiments/results.json) without being quoted in prose.

17.4 Sensitivity sweeps

Configured sweeps vary one configuration parameter at a time and report every arm against its reference. Summarizing those arms, the largest-magnitude paired mean effect is 0.252565 in dimensionless score units for the discount sweep, 0.120347 nats for the return-entropy bin-count sweep, and 0.757051 nats for the policy-temperature sweep. These are descriptive effect sizes of a synthetic computation under parameter changes; they measure the arithmetic sensitivity of the utilities, not any cognitive or empirical quantity.

17.5 Validity controls

The runner also executes analytic negative controls. The analytic-identity control suite passs (reported value 1), and the invalid-input control suite passs (reported value 1); each reported numeric value is the registry encoding of the corresponding control outcome, in which every analytic identity check and every malformed-input rejection behaved as declared. Separately, the crisp two-role POVM example satisfies completeness to a mean residual of 2.1684e-17 with interval [1.56478e-17, 2.77203e-17] and a maximum residual of 2.22045e-16 across replicates, consistent with the matrix contracts in sec. 19.

17.6 Aggregation and localization diagnostics

A final study mirrors the post-extraction aggregation and localization ablations of the Democritus publication [Mahadevan, 2026b] on a seeded synthetic paraphrase-family corpus. Claim records are generated in families whose members share the exact canonical cause, effect, relation, polarity, and regime while varying only qualifier phrasing, and selected families additionally straddle two regime labels with a polarity-flipped duplicate on each side. Localization then groups records under exact canonical keys, so a record with an unspecified regime forms its own class and never merges with a regime-labeled one.

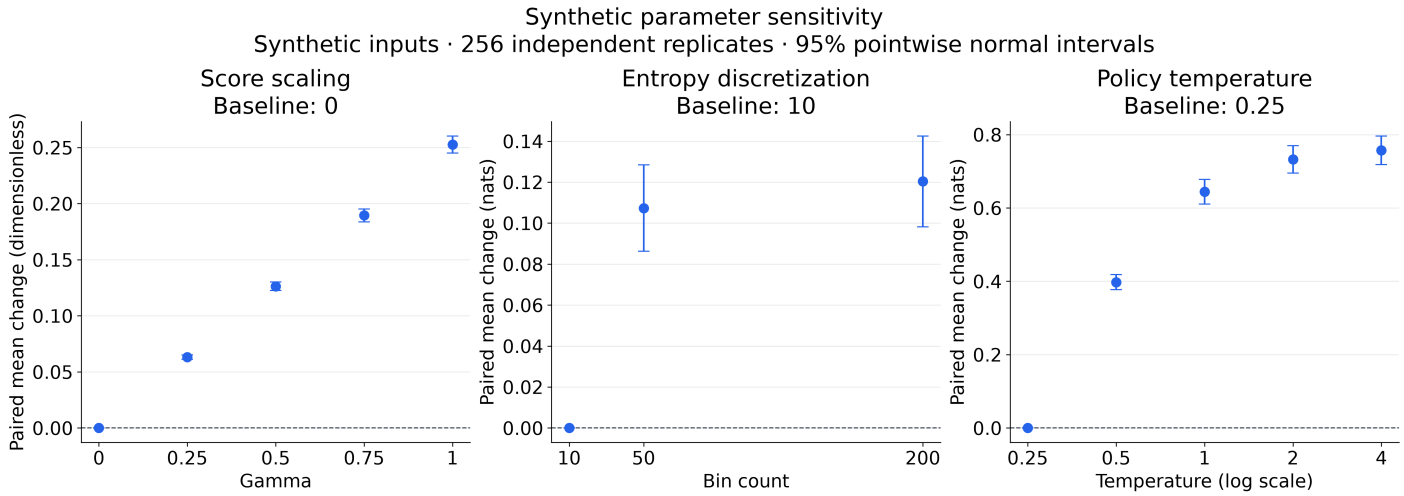


Figure 29: All configured sensitivity arms (discount, return-entropy bin count, policy temperature) plotted against their reference arms with pointwise paired intervals over seeded replicates. All effects are synthetic-computation sensitivities in dimensionless score units or nats; they carry no empirical interpretation.

Three contrasts are reported against this corpus, all at the replicate sample unit with the same normal-approximation intervals as the studies above. The no-localization contrast compares the number of localized classes with the raw record count: the mean inflation reduction is 0.498461 (dimensionless) with interval [0.477635, 0.519287]. The regime-projection ablation recounts polarity conflicts after projecting the regime away; the paired mean difference between the regime-resolved count and the projected count is 2.125 conflicts with interval [1.821, 2.429], where a positive value counts regime-resolved conflicts that collapse into merged ones under the projection. The closure diagnostic reports the mean triangle-closure ratio of the within-class compatibility graphs, 0.458591 (dimensionless) with interval [0.408272, 0.508909]; following the cited publication's own account of its compatibility complex, this ratio is a finite pairwise-closure statistic on observed pairs, not a horn-filling test in the nerve of a groupoid. Two descriptive counts complete the block without intervals: the regime-resolved polarity-conflict count 4.25 and the mean localized class count 14.375.

The boundary of this subsection is narrower than the rest of the section. Every corpus is synthetic, so no property of any real claim population is measured, and no real-corpus claim is made. The merge correctness of localization is unvalidated against any annotated standard; the cited publication itself reports such an independent post-extraction benchmark as its most important next step [Mahadevan, 2026b]. The numbers above validate the bookkeeping behavior of the aggregation and localization utilities on known synthetic constructions, nothing more.

17.7 Interpretation boundaries

Three boundaries delimit this section. First, replication intervals summarize variation across seeded runs of deterministic code; they are not sampling distributions over any population. Second, every configured arm is reported without multiplicity adjustment, and the configurations themselves are synthetic choices, not estimated models. Third, agreement with closed forms and numerical gradients validates the implementation of a stated computation; it neither extends the mathematics beyond its assumptions nor converts any utility into a Bellman solver, a general inference engine, or a measurement model. The reproduction path for every number in this section is documented in sec. 25.

18 Quantum Diagram Connections and Their Assumptions

Typed diagrams also occur in quantum information, where wires denote systems and boxes denote operations. This shared notation makes some structural comparisons possible. It does not identify a grammatical derivation, a quantum circuit, and a Bayesian update as the same process.

A categorical interpretation must specify its source category, target category, object map, arrow map, and preserved structure. The nonsymmetric ordering of pregroup grammar is especially important: a wire exchange available in a symmetric quantum-process category is not automatically a grammatical rewrite. A cup and a general ZX spider are also different operations; the presence of duality does not supply all the additional algebraic laws of a spider.

The local implementation uses DisCoPy diagrams and finite-dimensional NumPy matrices. It does not implement ZX rewriting, generalized-flow circuit extraction, a topological quantum neural network, a TQFT evaluation, a sheaf of semantic channels, or a hardware compiler. These are possible external frameworks for future work, not local validation results. No conclusion about quantum advantage, trainability at arbitrary scale, thermodynamic semantic capacity, or protection against prompt injection is drawn from their names.

The implemented measurement model in sec. 19 can be evaluated directly and is consequently a more concrete point of comparison. Even there, a probabilistic analogy does not imply that linguistic cognition uses physical quantum coherence. A useful next experiment would compare a specified measurement-based classifier against a matched classical model and report exactly which resources and assumptions differ.

19 Case Labels as Measurement Outcomes

A positive-operator-valued measure (POVM) is a family of positive semidefinite Hermitian effects E_c that sum to the identity. A density matrix ρ is positive semidefinite and Hermitian with trace one. Under these assumptions,

$$P(c \mid \rho) = \mathrm{Tr}(E_c \rho), \qquad \sum_c P(c \mid \rho) = 1. \tag{14}$$

The sum identity follows by linearity of trace. The implementation checks finite entries, dimensions, Hermiticity, positivity, completeness, and density normalization within explicit floating-point tolerances. A symmetric eigenvalue routine alone would not validate Hermiticity, which is checked separately. An individual effect is also required to lie below the identity.

`crisp_case_povm()` constructs orthogonal coordinate projectors. A projective measurement does not guarantee a deterministic outcome for every state: a mixture or superposition can yield several outcomes. The convenience `semantic_state()` constructs a diagonal mixture from supplied nonnegative role weights; it does not create coherence or entanglement.

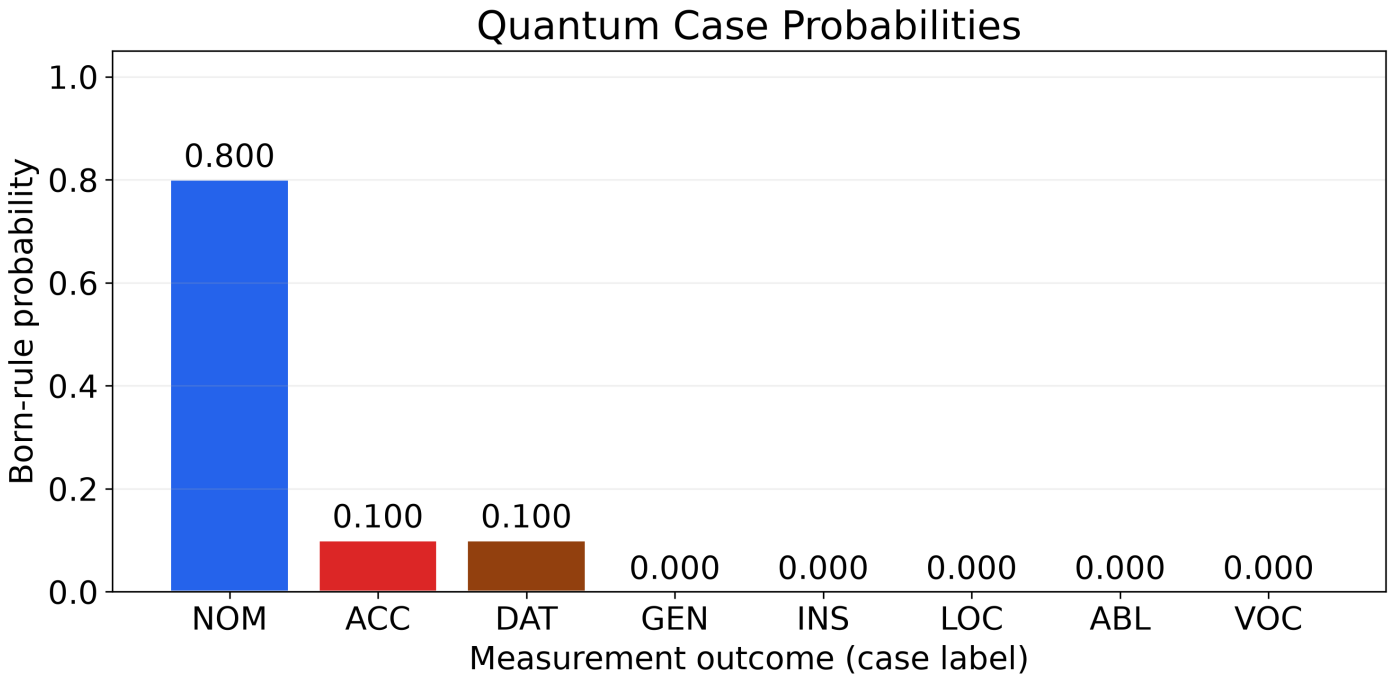


Figure 30: Born-rule probabilities for orthogonal projectors and the synthetic diagonal mixture, with computed outcome probabilities 0.8 on NOM, 0.1 on ACC, and 0.1 on DAT. Remaining outcomes have zero probability. These bars are computed traces; they show neither interference fringes nor a continuous semantic-state density.

`graded_case_povm()` creates diagonal effects from normalized role weights on basis coordinates. `fluid_s_povm()` rotates a two-role measurement according to a supplied context parameter. These are finite mathematical models. Interpreting their outcomes as case assignments is a chosen labeling convention, not a proof that grammatical alignment equals a quantum reference frame.

A general caller can provide a coherent density matrix directly, subject to validation. Demonstrating an interference effect would require a specified state family and measurement whose probabilities depend on off-diagonal components. The canonical figure uses commuting diagonal objects and admits an ordinary classical probability interpretation. No physical quantum advantage or semantic channel-capacity result is established. The seeded completeness check of the canonical example is reported in sec. 17.5.

20 Proposed Case-Labeled Agent Interfaces

Role labels can make an agent interaction easier to inspect: a requester initiates a task, an instrument executes an operation, an object is processed, and a recipient receives a result. A case-based interface could attach such labels to typed records and check allowed compositions. This is an engineering proposal, not a claim that existing agent protocols lack semantics or that linguistic case alone defines a secure protocol.

A useful record would include authenticated actor identity, operation type, resource identifier, authority scope, input provenance, and result destination. These fields must be distinguished from user-editable text. Calling a piece of content "NOM" cannot grant it authority.

Table 3: An illustrative interface vocabulary, not a deployed protocol.

Proposed label	Possible interface meaning	Required additional information
NOM	Initiating actor	Authenticated identity and authorization
INS	Tool or executor	Capability, constraints, and executable operation
ACC	Processed content	Origin, trust level, and allowed uses
DAT	Recipient	Destination authorization and disclosure scope
LOC / ABL	Context / source	Actual environment and provenance records

A composition check can establish that an output type is accepted by the next operation. It cannot by itself establish that either operation is truthful, authorized, privacy-preserving, or successful. The present repository supplies no network transport, protocol adapter, credential system, or reference monitor. A future prototype should connect the labels to real capability checks and evaluate both rejected unauthorized actions and permitted legitimate actions.

Entity histories from sec. 10 could support an interaction trace if identities were supplied reliably. They would still need explicit semantics for state updates, delegation, revocation, and failures. No fixed-point theorem for multi-agent behavior follows from representing a dialogue as a graph.

21 Role Policies and the Limits of Type-Based Security

A useful prompt-injection analogy is the attempted promotion of processed content into instructions with authority. The security module represents selected role transitions and reports violations relative to a supplied finite policy. It operates on already assigned labels; it does not read arbitrary text and infer trustworthy authority.

`detect_type_violation()` rejects roles outside its category before considering identity transitions. `CaseFrameValidator` checks membership and pairwise licensed connections, refreshing its adjacency view when the mutable category changes. The assignment checker accepts a connection in either direction for its pairwise compatibility test, so its result must not be mistaken for verification of a directed execution trace. Reported severities are hand-selected policy scores — fixed constants in the checker rather than tunable policy inputs — and not calibrated attack probabilities.

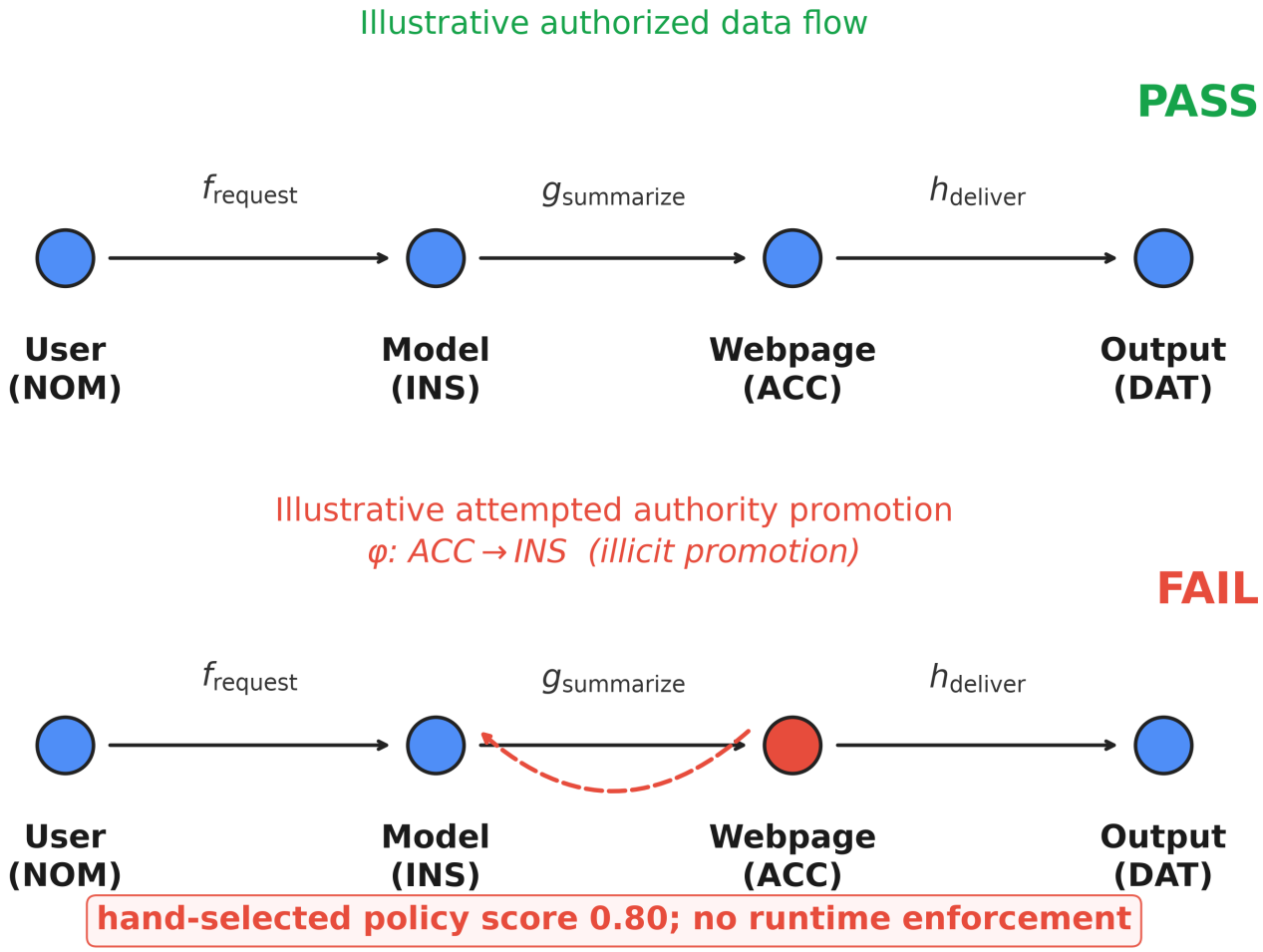


Figure 31: A schematic authorized flow and an attempted content-to-authority promotion. The drawing communicates a policy distinction; it is not an observed attack trace or an executable firewall.

The compatibility class `MonoidalFunctor` checks pairwise role separation and existence of corresponding edges. Its method name `preserves_tensor()` is historical: the implementation has no tensor objects or tensorator coherence data. Monoidal functors need not be injective on objects, and two roles mapping to one object does not establish nonmonoidality. Indeed, grammatical alignment deliberately merges some argument distinctions. Whether a merge is prohibited is a policy decision.

21.1 Requirements for operational enforcement

A deployed system would need authenticated provenance, trustworthy assignment of roles, authorization at the tool boundary, and a check that the executed action matches the checked request. It would also need adversarial tests spanning indirect instructions, multi-turn context, confused delegation, and permitted role changes. The repository implements none of those system boundaries and reports no detection rate or security benchmark.

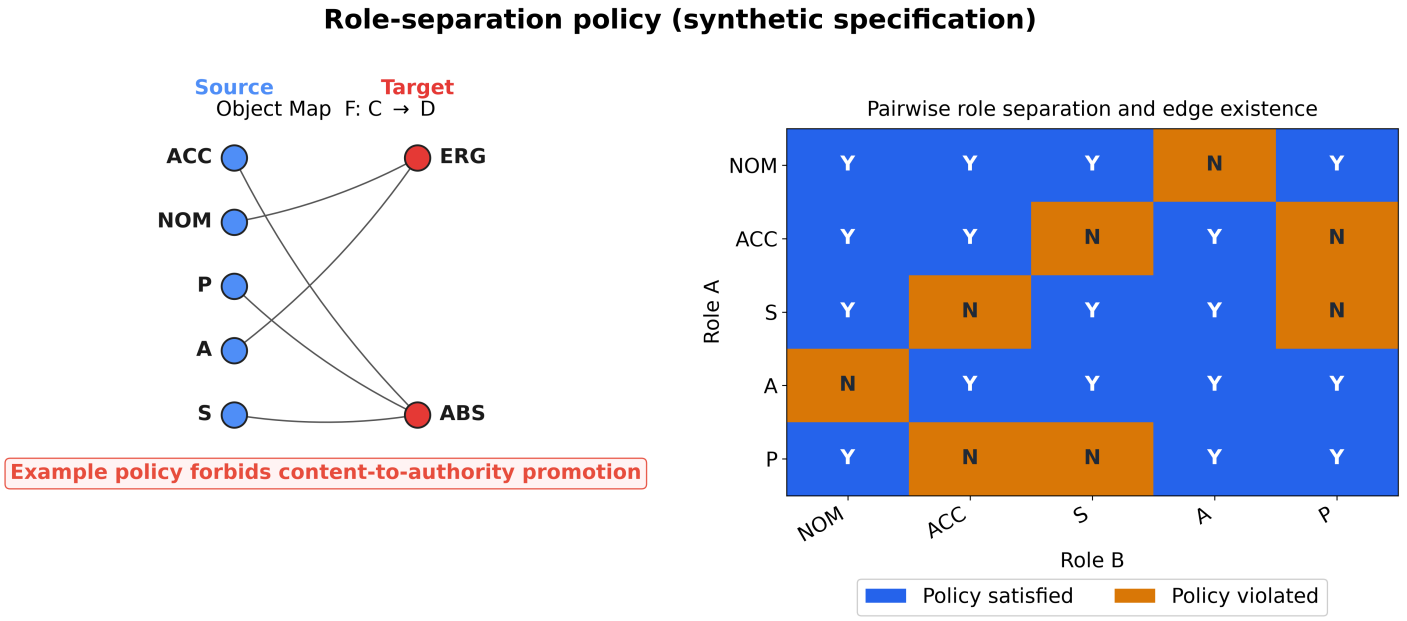


Figure 32: Policy cell matrix for the supplied role-separation rules: each cell records whether role separation and the required edge hold under this rule. A failed cell records a label merge or missing edge under this rule, not mathematical failure of a monoidal law and not evidence that a linguistic alignment is malicious.

21.2 Limits

Role labels alone cannot detect deception, validate factual claims, or prevent disclosure. A rejected label transition may be a legitimate request under a different policy; an accepted transition may carry harmful or false content. Magnitude, graph topology, and quantum terminology provide no automatic security guarantee. The finite checker remains useful as a transparent specification example when its inputs, rule set, and missing enforcement layer are stated.

22 Conclusion

Case diagrams provide a compact setting for comparing relational labels, typed composition, uncertainty, and interpretation. The executable examples show how a chosen representation can be built and inspected. They also expose where a proposed identification fails: a similarity table may violate enrichment, profile counts do not decide Morita equivalence, a finite score distribution is not a Bellman return law, and a type-policy result is not a deployed security guarantee.

The project's contribution is an evidence-bounded synthesis with explicit computational contracts. Pregroup reductions and tensor calculations are implemented examples. Role-history drawings are supplied bookkeeping. Numerical figures use synthetic inputs. The measurement example computes valid probabilities for specified matrices. Cognitive advantage, physiological interpretation, inter-theory transfer, and operational agent security remain hypotheses or engineering proposals.

22.1 Next experiments and proofs

Priority extensions follow from the missing evidence. A linguistic evaluation should specify a corpus and annotation scheme, estimate parameters on training data, and test held-out constructions against baselines. An enrichment study should report the raw matrix, axiom failures, closure changes, and sensitivity of magnitude. A topos bridge should supply actual theories and an equivalence witness. A cognitive or EEG study should prespecify outcomes and compare the model with simpler explanations. A protocol prototype should enforce authenticated capabilities and test the full execution boundary. Post-extraction aggregation adds one more: recent causal-discourse work states as its own most important next step an independent benchmark with expert annotations of merge, non-merge, and gluing decisions over paraphrastic, partially overlapping, regime-sensitive claim clusters [Mahadevan, 2026b]. The synthetic aggregation experiments in sec. 17 rehearse the decision structure on constructed inputs; the extension below asks whether those decisions survive expert annotation.

Extension	Required evidence	Falsification criterion
Linguistic evaluation	A specified corpus and annotation scheme; parameters estimated on training data; held-out constructions compared against baselines	Held-out performance is absent, not reproducible, or not better than the stated baseline
Enrichment study	Reported raw matrix, axiom failures, closure changes, and sensitivity of magnitude	Axiom failures persist after closure, or magnitude is insensitive to the declared structure
Topos bridge	Actual theories from the presentations and an equivalence witness	No witness is produced, or counts change under equivalent presentations while equivalence is still claimed
Cognitive or EEG study	Prespecified outcomes and comparison against simpler explanations	Effects disappear under the simpler explanation or the outcomes were not prespecified
Protocol prototype	Authenticated capabilities enforced across the full execution boundary	Execution proceeds without authenticated capabilities
Post-extraction aggregation	Expert-annotated merge, non-merge, and gluing decisions on paraphrastic, partially overlapping, regime-sensitive claim clusters	Annotated decisions disagree with the synthetic aggregation lane's decisions, or the agreement does not survive regime or polarity variation

Each extension should preserve counterexamples that can falsify its checks. Passing tests supports the stated software contracts; it neither supplies absent experiments nor proves all mathematical interpretations. Keeping these distinctions explicit makes further work reproducible and its results easier to assess.

23 Appendix A: Schematic Construction Panel

The panel collects eight hand-authored construction sketches. Its arcs are dependency-style annotations, not constituency trees produced by a parser. Some panels include a checked simple type template; the relative-clause and causative sketches display their interpretive structure without claiming a completed formal derivation.

The English examples use assigned NOM/ACC/DAT/INS labels to illustrate argument functions; those labels do not assert corresponding English case inflections. A benefit recipient or passive agent should not be equated with a particular morphological case across languages. Readers seeking executable reductions should use the DisCoPy constructors in sec. 6, whose types and codomains can be checked directly.

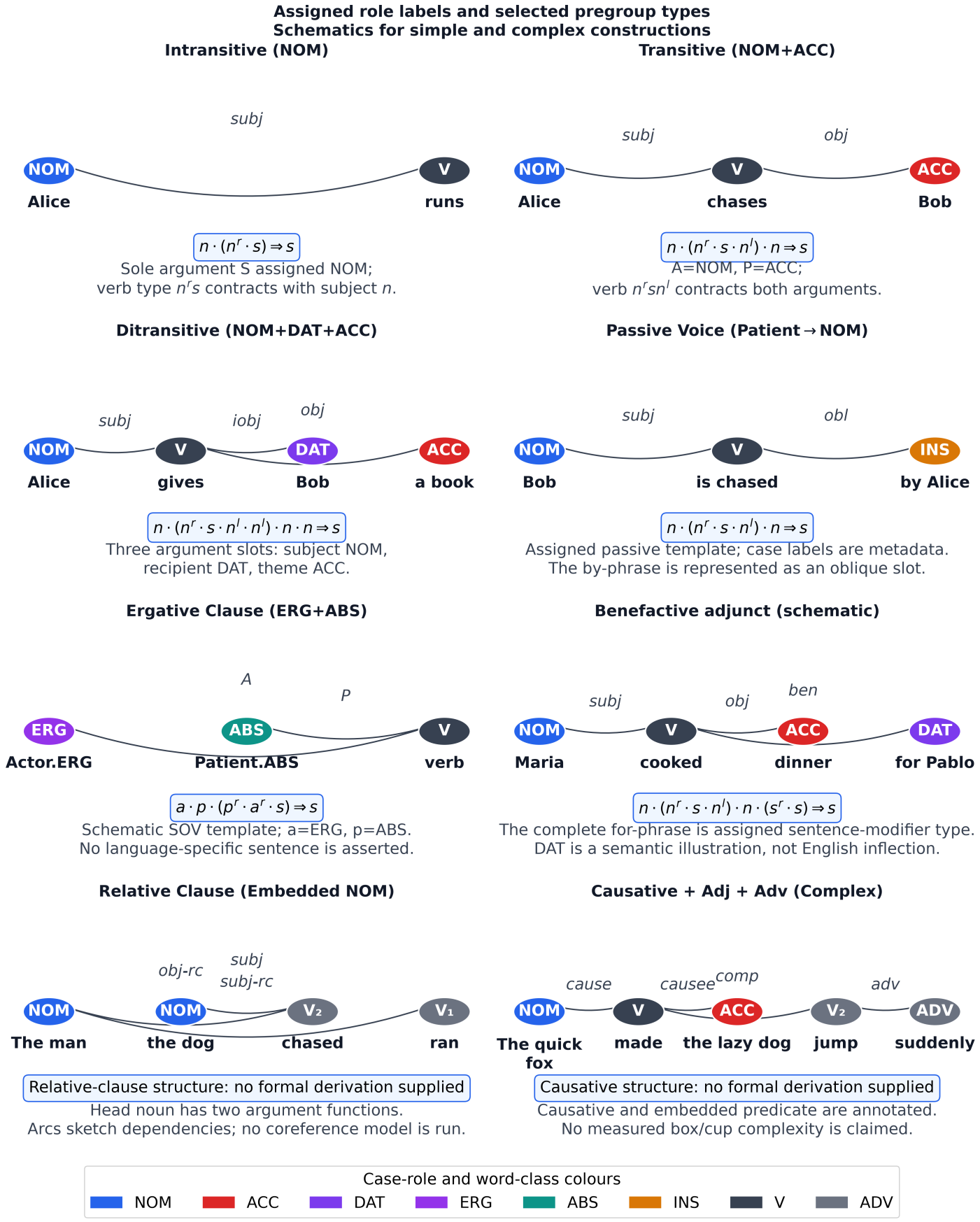


Figure 33: 8 pedagogical sketches spanning intransitive, transitive, ditransitive, passive, ergative, benefactive, relative-clause, and causative patterns. Words, roles, and arcs are supplied by the generator. The ergative example uses schematic labels rather than an unverified language-specific sentence. Panels are not ordered by a measured complexity score.

24 Appendix B: Notation and Conventions

24.1 Linguistic labels

S, A, and P denote the argument primitives defined in sec. 4. NOM, ACC, GEN, DAT, INS, LOC, ABL, VOC, ERG, and ABS are case labels used in the examples. Their interpretation depends on the language or modeling context. Case color is redundant with a written label; it does not encode evidence strength.

24.2 Algebra and probabilities

Table 4: Symbols used in the mathematical and computational examples.

Symbol	Meaning here
A, B, C	Objects or specified role labels
$f : A \to B$	A typed arrow; its exact semantics must be supplied
$g \circ f$	First f, then g
$\otimes$	Monoidal product in a specified category
n, s, n^l, n^r	Noun, sentence, and adjoint types
Z	Candidate or verified hom-value matrix (entries Z_{ij}) or similarity matrix, distinguished in context; in sec. 16.4 the scalar score random variable in $\mathrm{Var}(Z)$
$\lvert Z \rvert$	Weighting/coweighting sum when defined
q, p, L, T	Posterior, prior, likelihood, row-stochastic transition matrix
F	Variational free energy for a specified fixed model; in quantile-coverage contexts (sec. 17.2) the distribution function of the return law, as in $F(Q(\tau))$
G	Caller-defined policy score in the compatibility API
γ	Discount factor in $[0, 1]$ for the return distribution and policy score
β	Risk coefficient on $\mathrm{Var}(Z)$ in the caller-defined policy score
$\tau, Q(\tau)$	Quantile probability level and value
κ	Huber clip threshold in the quantile update; distinct from the complexity score $\kappa(D)$ of sec. 9
η	Distortion parameter; the stated distortion mode is optimistic iff $\eta < 1$
λ	Precision vectors (likelihood, prior, post) in the filtering and proxy formulas
m, m_0	Return-distribution mean and baseline return in the N400-inspired proxy
$N_{\mathrm{lex}}, N_{\mathrm{cup}}, N_{\mathrm{cap}}, d(D)$	Lexical item, cup, and cap counts and the complexity score $\kappa(D)$
E_c, ρ	POVM effect and density matrix
W_p	Wasserstein distance of order p under a stated reconstruction
DPE	Distributional prediction error (surprisal or Wasserstein form, as stated)

Natural logarithms give entropy and KL in nats. Similarity weights and synthetic scores are dimensionless unless a scale is explicitly supplied. Variances have squared score units. ERP-inspired amplitudes are uncalibrated model units; waveform time coordinates are milliseconds. `return`, `Bellman`, `Bethe`, `MonoidalFunctor`, and `ClassifyingTopos` in legacy APIs do not override the narrower operational definitions in the text.

Replication intervals in the synthetic-statistics chapter (sec. 17) are simulation-replication summaries of seeded runs, not sampling distributions over any population.

25 Appendix C: Reproduction and Verification Scope

Two different counts matter and must not be conflated. The suite *collects* 1909 tests in 94 files across 12 source subpackages; a collection count alone certifies nothing. A passing run is certified only by the source-bound quality receipt, which records 1909 passed, 0 failed, and 0 skipped tests with 7090 of 7549 statements covered (92.44% combined). The receipt binds that evidence to source fingerprint c9f11b706217 and was recorded at 2026-11-11T08:32:14.687229+00:00; a receipt whose fingerprint no longer matches the tree is rejected. The metrics writer records installed NumPy 2.5.4 and DisCoPy 1.2.2.

The suite covers role-graph construction; sampled endpoint and weight laws; explicit pregroup reductions; tensor evaluation; finite probability calculations; quantile update conventions; matrix magnitude and composition checks; POVM and density validation; finite role-policy checks; the seeded synthetic studies of sec. 17; figure generation; and manuscript substitution. It uses real numerical arrays, files, and subprocesses. Test counts, coverage, and synthetic study outputs are not mathematical proofs or empirical effect sizes.

Negative controls include malformed probability inputs, impossible observations, non-Hermitian effects, incompatible quantile grids, a composition-violating matrix, a matching theory profile without a transfer witness, manuscript math containing literal dollar delimiters, unregistered or nonfinite manuscript variables, and prose or captions containing numeric literals that no registry identifier backs. These cases matter because a validator that accepts everything can produce a superficially green run.

From the project root, install and verify with:

```
uv sync --frozen --extra mcp
uv run python scripts/quality_gate.py --coverage
uv run python scripts/run_experiments.py
uv run python scripts/generate_diagrams.py
uv run python scripts/inject_variables.py
uv run python scripts/validate_project.py
```

All canonical numerical examples are synthetic. Seeds or fixed arrays are declared in source. The project has no corpus split, participant sample, EEG dataset, quantum-hardware receipt, or operational-agent security evaluation. This revision is identified by 10.5281/zenodo.22774403; the series uses DOI 10.5281/zenodo.19695259. The live deposit state is reported at the version-record link.

The first command uses the lockfile; the quality gate runs lint, types, and the full coverage-enforced suite and writes the receipt under `output/reports/`. The experiments runner writes the schema-annotated synthetic results consumed by injection. Figure generation writes `../figures/` and the figure registry, binding each entry to the source fingerprint it was rendered from. Injection writes `output/manuscript/` and the variable manifest from the canonical numbered sources; validation re-checks every binding. Render through the sibling template engine using the lifecycle-qualified project selector documented in the repository README. Rendered PDFs require visual inspection in addition to text, link, and citation checks.

References

Abdullah Akgül, Gulcin Baykal, Manuel Haußmann, Mustafa Mert Çelikok, and Melih Kandemir. Distributional active inference, 2026. URL https://arxiv.org/abs/2601.20985.

Tai-Danae Bradley, John Terilla, and Yiannis Vlassopoulos. An enriched category theory of language: from syntax to semantics. *arXiv preprint arXiv:2106.07890*, 2021. URL https://arxiv.org/abs/2106.07890.

Olivia Caramello. The theory of topos-theoretic 'bridges': a conceptual introduction. *Glass Bead Journal*, 2016. URL https://www.glass-bead.org/article/the-theory-of-topos-theoretic-bridges-a-conceptual-introduction/.

Olivia Caramello. *Theories, Sites, Toposes: Relating and Studying Mathematical Theories through topos-theoretic 'bridges'*. Oxford University Press, Oxford, 2018. ISBN 9780198758914. doi: 10.1093/oso/9780198758914.001.0001.

Bob Coecke, Mehrnoosh Sadrzadeh, and Stephen Clark. Mathematical foundations for a compositional distributional model of meaning. *Linguistic Analysis*, 36:345–384, 2010.

Bernard Comrie. Alignment of case marking of full noun phrases. In Matthew S. Dryer and Martin Haspelmath, editors, *The World Atlas of Language Structures Online*. Max Planck Institute for Evolutionary Anthropology, Leipzig, 2013a. URL https://wals.info/chapter/98.

Bernard Comrie. Alignment of case marking of pronouns. In Matthew S. Dryer and Martin Haspelmath, editors, *The World Atlas of Language Structures Online*. Max Planck Institute for Evolutionary Anthropology, Leipzig, 2013b. URL https://wals.info/chapter/99.

Will Dabney, Georg Ostrovski, David Silver, and Rémi Munos. Implicit quantile networks for distributional reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1096–1105, 2018a. URL https://proceedings.mlr.press/v80/dabney18a.html.

Will Dabney, Mark Rowland, Marc G. Bellemare, and Rémi Munos. Distributional reinforcement learning with quantile regression. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence (AAAI-18)*, pages 2892–2901, 2018b. doi: 10.1609/aaai.v32i1.11791.

Karl J. Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. Active inference: a process theory. *Neural Computation*, 29(1):1–49, 2017. doi: 10.1162/NECO_a_00912.

Tom Leinster and Michael Shulman. Magnitude homology of enriched categories and metric spaces. *Algebraic & Geometric Topology*, 21(5):2175–2221, 2021. doi: 10.2140/agt.2021.21.2175.

Sridhar Mahadevan. Categories for AGI. Draft digital edition, 2026a. URL https://categorical-ai.sridharmahadevan.com/books/categories-for-agi.

Sridhar Mahadevan. Democritus: Homotopy-localized causal discourse extraction from language. *Entropy*, 28(9):986, 2026b. doi: 10.3390/e28090986.

Anna Siewierska. Alignment of verbal person marking. In Matthew S. Dryer and Martin Haspelmath, editors, *The World Atlas of Language Structures Online*. Max Planck Institute for Evolutionary Anthropology, Leipzig, 2013. URL https://wals.info/chapter/100.